

Block**2****HEALTH AND HYGIENE ISSUES IN DEVELOPMENT
JOURNALISM -II**

UNIT 5
Health and Sanitation

UNIT 6
Education and Media

UNIT 7
Media and Environment

UNIT 8
Economy and Finance

UNIT 9
Industrialisation and Urbanisation



UNIT 5 HEALTH AND SANITATION

Structure

- 5.0 Introduction
- 5.1 Learning Outcomes
- 5.2 Health and Sanitation in India: Overview
- 5.3 Health and Sanitation: Existing situation in India and Other developing Countries
- 5.4 Sanitation, Quality Use, Access and Trends (SQUAT) Survey
- 5.5 Health and Sanitation: Consequences among women and Children
- 5.6 Health and Sanitation: Sociological Understanding
- 5.7 Media, Health and Sanitation
- 5.8 Let Us Sum Up
- 5.9 Glossary
- 5.10 References and Further Readings
- 5.11 Answer to Check Your Progress Exercises

5.0 INTRODUCTION

Health and Sanitation are interrelated concepts. As we all know, maintaining excellent and proper Sanitation will safeguard human health. It is an irrefutable fact. It is the responsibility of all stakeholders for ensuring accessible and affordable sanitation for all. Adequate health and sanitation are essential for human dignity, mostly women and children's dignity. There are various measures at the international and national level to provide affordable and accessible sanitation facilities to all. Governments would have achieved Goal 3 to 7 of Millennium Development Goals (MDGs) if they had provided sanitation facilities to all. Despite the targets fixed in MDGS, nearly 2.3 billion people didn't have access to improved sanitation facilities and 892 million people still defecate open. Goal six of Sustainable Development Goals (SDGs) talk about 'Ensure availability and Sustainable Management of water and sanitation for all'. Target 6.2 aims to "achieve access to adequate and equitable sanitation and hygiene for all and end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations" It also targets to achieve goal six by 2030. International Organisations like The world Bank, United Nations played a significant role in highlighting the significance of basic amenities at households, social infrastructure to address the adequate standard of living, right to health and other human rights.

Article 25 of The Universal declaration of human rights states "Everyone has the right to a standard of living adequate for the health and well-being of himself and for his family, including food, clothing, housing and medical care and necessary social services... (UN 1949)".

We have given data about the open defecation in this Unit. We have also discussed the impact of lack of Sanitation among women and children and the responsibilities of media to achieve 100 per cent Open Defecation Free (ODF) in India. Let us start looking at the overview of health and Sanitation in India.

5.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- explain the overview of health and sanitation in India;
- examine the situation of sanitation in India with other countries;
- describe the impact of Open defecation among women and children; and
- examine the role of media in improving the health and sanitation among all.

5.2 HEALTH AND SANITATION IN INDIA: OVERVIEW

We will recognize the significance of proper health and sanitation among human beings and improve the same by looking at the existing health data. Prevalence of open defecation has direct impacts on the health of human beings. Open defecation contaminates water, leading to the spread of many diseases and subsequent death of human beings. The study found that 88 per cent of diarrhoeal diseases are due to unsafe water, inadequate sanitation and hygiene. Diarrhoeal disease (including cholera) kills 1.8 million people every year, including 90 per cent of children under five. This is happening mostly in developing countries (Reddy, Sudhakar, 2016).

73rd Amendment to the Constitution (243G of Eleventh Schedule) assigns rural housing, sanitation, and drinking water inclusive of 29 duties to Panchayats. In the beginning, the concept of sanitation denotes the disposal of human excreta. The concept widened its meaning, and presently the sanitation means disposing of human excreta. It means the disposal of solid and liquid waste disposal and management, personal hygiene, food hygiene, personal hygiene, domestic and environmental hygiene. Proper sanitation contributes to a good social life. Sanitation is fundamental determinants of quality of life and human development index (The definition of quality of life and human development index are part of the Unit “Women and Child Development). The proper disposal of human excreta safeguards the water and soil quality. Hence, we can prevent many diseases. Thus, the concept of sanitation includes personal hygiene, home sanitation, safe water, garbage disposal, excreta disposal, and wastewater disposal (Reddy, Sudhakar, 2016).

According to the World Bank’s World Development Indicators, 2012 forty-three out of every 1000 babies born in India died before their first birthdays. The World Development Report 2012 further indicates that they collected data from 194 countries. Out of these 194 countries, Infant Mortality Rate (IMR) of 150 countries is lower than India. When we compare India’s IMR with BRICS (Brazil, Russia, India, China and South Africa) nations, it is three times higher than China, Brazil and Russia. Bangladesh, Kenya and Rwanda reported lower IMR rate than India. India’s IMR is only on par with Uganda and Myanmar (Coffey and Spears, 2017). Another significant indicator talks about the health of women and children are Maternal Mortality Rate (MMR). MMR (women dying during child birth per one lakh live births) in India was 556 in 1990. It declined to 254 in 2006-2014 and further decline to 130 in 2014-16. We will elaborate further in economic terms. International agencies predict that India’s IMR is 20 per cent higher than other countries with the same GDP as India. Usually, we assume

that poverty is the main reason for poor nutrition, high IMR, and high MMR among Indian women. Poverty is not only caused for poor health among women and children. But, poor sanitation, along with poverty, also makes India's newborn children and women vulnerable. There is no direct correlation between economic development and providing sanitation facilities. For example, the open defecation rate (household latrine non-ownership rate) according to the 2011 census is 67 per cent and 76.8 per cent respectively in the economically developed and well-governed states like Gujarat, Tamil Nadu. The open defecation rate in Gujarat and Tamil Nadu can be comparable with poor states like Bihar (82.4 per cent) and Uttar Pradesh (78.2 per cent). But, Northeastern states have lower rates of open defecation (Coffey and Spears, 2017). We explained this phenomenon further in this Unit later through a case study. According to 2011 census, open defecation in Sikkim is 15.9 per cent, 14.0 per cent in Manipur, and 15.4 per cent in Mizoram.

You must be wondering why we are talking about open defecation in the Unit Health and Sanitation. Before we know the reasons, we need to see the definition of open defecation. The term 'Open Defecation' is used throughout the world for throwing faeces in the open field, roadside, agriculture field, forest or any available space. It also means that relieving nature calls by sitting squat position on the open ground. The relieved human excreta contains faecal germs will infect other human beings which causes diarrhoea. The International Organisations UNICEF and World Health Organisation (WHO) regularly collect data on open defecation and publish. These organisations' joint monitoring report estimates that 13 per cent of the world population disposes of its faeces in open. The other 87 per cent disposes of excreta using some latrine or toilet. It ranges from the expensive comfortable toilet with double septic tanks to pit latrine. However, people in many countries use open defecation due to a range of reasons like lack of water facilities, poverty, lack of access, inability to afford one etc. The issue of open defecation and providing improved sanitation facilities for all in India is challenging and is a unique problem. Learners will recognize the reasons for mentioning as unique as far as India is concerned. According to the 2011 census, 53 per cent of Indian households had no toilet and did not use the public toilet. This figure is higher in rural households than in urban homes. 13 per cent of urban home lack a latrine or toilet compared to 70 per cent of rural households. Thus the majority of open defecation happens in rural India. Compared to the 2001 and 2011 census, the rate of declining open defecation is very slow. Open defecation is falling one percentage point per year in both rural and urban area (Dhaske and Parasuraman, 2016 and Coffey and Spears, 2017).

In India, Thirty-two per cent defecate open having bachelor degree and above. Fifty-two per cent adult in the household having educated and completed secondary school defecate open.

Government of India started implementing sanitation programmes since 1986. Government of India formulates plans and policies related to sanitation and implemented through State governments and Local body institutions. It all started through the Central Rural Sanitation Programme (CRSP) in 1986 to construct subsidized latrines to the Below Poverty Line (BPL) families. Along with toilet construction, the programme also had objectives to create awareness among people on toilets usage, converting dry latrines to pour flush latrines and constructing community toilets. At the beginning of the five-year plan period, the programmes/schemes were designed and implemented as supply drive mode,

and most of the plans were formulated based on the top-down approach. After realizing the significance of community participation, the government started perceiving policies and programmes through a bottom-up approach. Based on the bottom-up approach, the existing Central Rural Sanitation Programme (CRSP) programme has converted and implemented as the Total Sanitation Campaign (TSC) in 1999. Government formulated TSC as the demand-driven programme. Households need to request the government for financial assistance to built toilets.

Along with the demand-driven approach, it encouraged local governments to participate actively, and the government has announced the cash prize for local governments to successfully eliminate open defecation. Some local governments could able to achieve the elimination of open defecation. Still, the programme is not as successful as expected. Experts carried out the evaluation study. According to Andres Huesco, Sanitation expert, the Percentage of households owns latrine has gone only nine percentile points from 2001 to 2011 by comparing the data from 2001 and 2011 census about TSC. After that, in 2012, the government has launched Nirmal Bharat Abhiyan to construct more toilets. All these programmes emphasized the construction of toilets for individual households and community through various modes like a subsidy. To further accelerate the universalization of toilet coverage, the Indian government launched a Swachh Bharat Mission on 2nd October 2014 to make India clean by 2019.

5.3 HEALTH AND SANITATION: EXISTING SITUATION IN INDIA AND OTHER DEVELOPING COUNTRIES

According to the WHO, 2.6 billion people of 42 percent of world population lack access to adequate sanitation in the world (The Hindu, October, 26, 2008). Two out of every three among south Asia's 1.5 billion population lack appropriate sanitation facilities (SACOSAN III.2008). We think that economic progress may bring health and sanitation for all. When we compare the data of economic progress and sanitation facilities' availability among human beings, it surprises us. World Bank's International Comparison Programme computes India's Gross National Income per capita. India's Gross National Income per capita grew at an average of 8 per cent per year between 2001 and 2011. India's Income grew 46 per cent higher than sub-Saharan Africa and 56 per cent faster than the rest of the world. However, the population using toilets in India fall behind the rest of the world. In neighbouring Bangladesh, the demographic and health survey (DHS) finds the latrine use of rural parts of the country is almost universal in the last twenty years increasing from 67 per cent of households in 1994 to over 95 per cent in 2014. India has a higher proportion of population defecate open filed than any other developing country. Ninety per cent of poorer countries than India also have lower rates of open defecation (Spears and Coffey, 2017).

Bangladesh, as a country, suffered a lot due to open defecation. It was a birthplace of first cholera (disease spread by faecal pathogens that leads to swift death due to dehydration) pandemic. By looking at the various surveys in Bangladesh, open defecation has drastically decreased, and less than 5 per cent of households defecate open as per the 2014 survey.

Secondly, we think that poverty is the leading causes of open defecation. We can dismiss this argument by looking at other countries data, especially with

neighbouring Bangladesh. In the year 2006, 52 per cent of Bangladesh households had dirt floors and no electricity. In 2005, 21 per cent of households had dirt floors and no electricity in India. Among impoverished households in Bangladesh, only 28 per cent defecate open, and 84 per cent of households in India defecate open. This data further open up our understanding. A simple latrine is not expensive. However, we need to find out the reasons behind open defecation and address the same (Spears and Coffey, 2017).

The design of the toilet in Africa is simple. Many latrines in rural Africa do not have a water seal. They do not need to flush. The same design can be done in India even though India has enough water sources close their home and living locality. According to the 2011 census, half of the household premises with water do not have the latrine in their home. We compared India with other countries with water sources and economic development. Next, we can compare India with other countries with an education. We assume that educated people do not defecate open. The adult literacy rate of 28 countries in the world is lesser than India. Out of these 28 countries, open defecation of 23 countries is less than India. Thus education is not the main culprit for open defecation. Another critical indicator we can compare for open defecation is governance. According to political Scientists, Monty G. Marshal and Ted Robert Gurr, the governance in countries like Afghanistan, the Democratic Republic of Congo, Haiti, Liberia, Myanmar, Pakistan and Sierra Leone are worst than India.

Nevertheless, significantly less percentage of people only defecate open in the countries as mentioned above. According to the World Bank easy doing business Index, It is challenging to do business in 47 countries than India. Out these forty-seven countries, thirty-nine countries (83 per cent) have a lower rate of open defecation than India (Spears and Coffey, 2017).

5.4 SANITATION, QUALITY, USE, ACCESS AND TRENDS (SQUAT) SURVEY

This survey is unique, unlike other surveys that try to determine the percentage of people having access to toilet facilities and reasons for not using toilets even though it is available on their premises. SQUAT survey to find out the reason for not using toilets through different dimensions. Even if the government provides toilets for everyone, it is the waste of resources if the people are not willing to use. The government needs to explore the reasons for not using toilets even though the government is willing to construct toilets for each household free of cost.

Dianne Coffey and Dean Spears and their team started their SQUAT survey in 2013 to study sanitation quality, use, access and trends in rural north India. They travelled thirteen districts of five States namely Bihar Madhya Pradesh, Rajasthan and Uttar Pradesh. According to their survey and their analysis, they found that these States are home for 30 per cent of the world population who defecate open. According to the 2011 census, 80 per cent of the population who does not own latrine belong to these States. They further found that open defecation is common among people who own latrines. The survey further found that 18 per cent of households surveyed, some people in the households use toilets and some defecate open. Forty per cent of households have working latrines in their sample, at least one member in the household defecate open. The SQUAT survey clearly shows

that open defecation is not about not having access to the toilet. The SQUAT survey points out that even if the government constructs latrines for each household, it is challenging to eliminate open defecation. This is due to the attitude of the people. Many could be able to afford latrine, but they fail to use the same.

5.5 HEALTH AND SANITATION: CONSEQUENCES AMONG WOMEN AND CHILDREN

Women and children specifically suffer in various ways for practising open defecation and lack of sanitation facilities in their home. It impacts their health, and they may face violence. Let us see this elaborately. More research related to the impact of open defecation on women and children's health and infectious diseases on human beings is underway rather than other aspects of open defecation (Cairncross S, Hunt C, Boisson S, Bostoen K, Curtis V. 2010). Suppose women and children get infected with human excreta which contains several harmful organisms mentioned earlier in this Unit, they face several health problems. According to Feachem RG, Bradley DJ, Garelick H, Mara DD in 1983, one gram of infected human excreta contains various microbes, including 106 pathogenic viruses and infectious virions, 106–108 bacterial pathogens, 103 protozoan cysts and 10–104 helminth eggs. Now we can understand the severity of the problem. If we inappropriately dispose of the human excreta in an open space, it poses significant health risks among human beings, mostly women and children. Here, we need to understand the reasons for talking about women and children specifically. Women and children tend to defecate the same place again and again for safety and security. It will lead to getting infected easily than male members in society. Human beings get infected with diarrhoea, typhoid, cholera, and viral infections from the human excreta's pathogens inappropriately. According to WHO, 1.8 million people in low and middle-income countries suffer from severe trachoma, which causes visual impairment (WHO, 2017). Human beings get infected with trachoma through flies that use human excreta as bread. It transmitted disease when it came in contact with human beings after using human excreta. The disease will spread from one person to another through eye discharge. Like trachoma, we need to take a note of another disease *schistosomiasis* (*snail fever*). There is an estimated 200 million people get infected with snail fever. This disease spread through human faeces to the freshwater snail. If the skin of the human being gets in contact with an infected snail, the disease will transmit. Human beings' immune system gets affected if human beings drink contaminated water and suffer from snail disease.

Everyone is in health risk due to open defecation. The risk among women and children would be severe. They face violence along with health risk. According to Strunz et al., Women with inadequate sanitation facilities are more susceptible to hookworm, which leads to anaemia among pregnant women. Anaemic pregnant women face the risk of delivering anaemic infant, and they face risk during pregnancy. Apart from this, Women and children face diarrhoea, malnutrition. If women face malnutrition, diarrhoea and pregnancy-related complications during reproductive age, the newborn child will face the same risk as women. Poor nutrition and open defecation may lead to low Body mass index (BMI) and low haemoglobin level among women. Low haemoglobin level during pregnancy creates complications during baby delivery, which is the main reason for high

MMR and IMR. This is the vicious cycle. We have explained in details about this vicious circle as Life Cycle Approach in the Unit “Women and Child Development”. Apart from the diseases mentioned above, open defecation increases vulnerabilities among women. Women tend to walk long distance to defecate, and they need to wait till sunsets to relieve their nature calls or do it before sunrise for want of privacy. Women may face physical abuse and sexual abuse when they walk long distances searching for a private place to defecate. Women face difficult hardship during the menstrual time.

The survey carried out by the Greenland et al. explained that children in the rural settlement of India engaged in open defecation are more susceptible to soil-transmitted helminths, an intestinal infection transmitted through the exposure of infective human faeces. The collected samples from the girls between the ages of 4 and 17. The Study further explained that older girls (over 12 years) were more likely to be infected with soil-transmitted helminths than younger girls.

Lastly, Kotian et al. found that people used open defecation in Bihar in India is more prone to get infected with the parasite. Moreover, they also observed that the infection was more prevalent in the female population (17.07%) than men (8.33%). They have also cited the reasons for the prevalence of infection in the study area. These include the low quality of available drinking water, the larger percentage of women engaged in livestock and agricultural management. Inappropriate waste disposal practices or other environmental conditions are also the main reasons. According to them, open defecation remains a significant cause of water contamination, spreading communicable diseases. Open defecation is the primary threat to public health.

However, the findings cannot be generalized to a larger population because they restricted the sample only among the patients admitted to the hospital.

The Study carried out by the Sahoo et al., and Hirve S et al. clearly shows that open defecation threatened women's dignity and privacy. They found the reasons for making women vulnerable due to open defecation from the selected sample by conducting group discussion. Women shared that they need to find a suitable place to defecate open. Women felt uncomfortable if male members cross that place, and it affects their privacy and dignity. The Study further found through the in-depth interview that women feared watched while defecating open. The Study listed the variety of psychosocial stress due to open defecation among women. These include; environmental (limited access, discomfort at defecation site, animals/insects) social (privacy, social restrictions and conflicts) and sexual stressors (peeping and sexual assaults). The most emphasized stressors were: searching for appropriate sites to defecate, travelling long distances, carrying water for cleaning, increased risk of insect or snake bites, fear of ghosts in dark and uncleanness at the site. The Study further reported that women prefer to travel in a group or go to the site with relatives for safety and security and avoid verbal or sexual abuse. Rural women face lots of psychological stress too. Hirve et al. study revealed that psychosocial stress extends to concerns regarding personal safety in rural western India. 64 per cent of respondents revealed in western parts of rural India that psychological stressors were the leading causes behind women feeling tensed, worried, depressed and irritated. Psychological stress has severe impact among girls and women of reproductive age as they face an additional challenge of managing their menstruation while tackling the

everyday need to defecation. However, Sahoo et al. argue that the issue extends beyond young age and is significant throughout all women's life stages. It is imperative to acknowledge that women's involvement in designing and placement of toilets stand the best chance of long-term success of reducing sanitation-related psychosocial stresses among women.

We need to acknowledge that Adolescent girls were more vulnerable to sexual harassment and assault. They need privacy and dignity to manage menstruation, and they need to maintain hygiene throughout their life to live healthily. Along with adolescents, newly married women, pregnant women, older women and children need proper sanitation facilities to maintain health. Open defecation directly impacts the height of the children. Stunting among children is common in India. Open defecation certainly impacts the ability of children learning. As we have seen in this Unit, Open defecation cause diarrhea and other water borne diseases among children. If children infect diseases regularly, children may not be able to attend the school regularly. It will certainly impact the ability of the children to learn. ASER study points out that children less exposed to open defecation fair better in learning. Women need accessible, affordable and clean toilet facilities for safety and security. The government formulated various schemes to construct individual toilets. Most government schemes failed to convince people to construct and use due to the lack of understanding of their health and hygiene. Many times government fail to engage women for policy making and implementation process. The existing gender relations in the households and patriarchy make it difficult for women to decide sanitation facilities at the household level.

5.6 HEALTH AND SANITATION: SOCIOLOGICAL UNDERSTANDING

As we have seen in this Unit, Government of India implements various programmes to construct household toilets and community toilets. Still, a considerable percentage of Indian families refuse to construct the toilets, and even if they construct, they fail to use the same. We need to unravel this puzzle to make India ODF.

As far as sociologists and anthropologists are concerned, Purity and pollution are part of the Indian socio-cultural aspects. Prof. M.N.Srinivas, in his book "Remembered Village", Prof. M.N.Srinivas stated that Indian Children taught to follow 'Purity and Pollution; at the early age and elders may do strict vigil on them to monitor their behaviours. For example, children taught to use the only right hand for eating and other auspicious purposes because they use the left hand to clean the private parts. Like that RS Khare note, villagers are more concerned about maintaining ritual purity than maintaining physical and environmental cleanliness. According to Diane, he often watched Uttar Pradesh, where parents hit their children when they take food by themselves through the left hand. We may think that ritual Purity is one reason people are not in favour of constructing toilets inside their home. They do rituals related to religion inside the home. Constructing toilets inside the home may pollute their rituals. Villagers in India consider childbirth and menstruation are polluting aspects. If a girl menstruates the first time, family and community allow her to mingle everyone after doing certain rituals after certain days. They do ritual after childbirth after few months before everything becomes normal.

Like that Sociologist, Damaris Luthi had studied hygiene behaviour in kottar in Tamil Nadu. As mentioned in the book *Where India Goes: by Diane Coffey and Dean Spears*, “he describes how Hindu value system the cleanliness and Purity is important. Purity in the home signifies as a reflection of one’s character and status. Many purity rules focus on homes. However, Luthi observes interest in cleanliness stops at the doorsteps of private homes. Furthermore, habits outside the homes are irrelevant and rubbish stamp”.

Patriarchy and superiority complex works in the minds of the Indians. This is the main reasons Indian cities become garbage dump. People literally through the garbage at the roads thinking that someone will come and clean the roads. Purity, pollution, caste and social subordination are the main reasons for low adaptation of latrines by the Indians villagers rather than poverty, availability, accessibility. When the government implemented Swacch Bharat Mission, the prime minister talked about the significance of Sanitation and Cleanliness. Every market and public places carry the board and request the people to throw garbage in the dustbin. Irrespective of notice board and leaders repeated request people still throw garbage on the roadside and around the garbage bin. The action of the people clearly shows the attitude of the Indians. Unless until they change the patriarchal attitudes and shed away the caste identity, it is difficult to bring the changes in the people’s mindset. According to Dhaske and Parasuram (2016), Young women do not prefer to use government constructed toilets due to cultural unacceptability and in-appropriate design. In India’s rural areas, many already constructed toilets are kept unused due to inadequate knowledge of sanitation. Their study further revealed that the community is not seeing sanitation is an essential component in their life. The existing government programmes lack the following areas, and these need to be fixed according to their study. These include the development of location-specific design and development of appropriate technology, Sensitising people on the significance of using proper sanitation for maintaining health, creating conscious among people on sanitation and cultural and gender-sensitive discourse on sanitation policy.

5.7 MEDIA, HEALTH AND SANITATION

Media writes ‘ Swacch Bharat Mission launched and Indian cities become clean, India is lagging behind the United Nations (UN) HDI indicators. Lack of water impedes people not to use toilets’. Most of the stories media fail to understand the real reason for open defecation and its consequences among women and children and the overall health of Indians. As we have already seen in this Unit, the present Swacch Bharat Mission made water storage tank available and public utilities after learning the lessons from previous Nirmal Bharat Abhiyan. India cannot quote the water storage facility and poverty for the everyday use of toilets and open defecation. WHO and UNICEF joint monitoring project on water and sanitation finds that 90 per cent of rural people in India have access to ‘improved water source’ which includes piped water, public taps and hand pumps tube wells and dug wells.

Compared to India, only 49 per cent of the people have access to improved water sources and 39 per cent of people rural people in sub-Saharan Africa defecate open. The media need to bring authentic data sources to make the community realise the significance of using toilets instead of not using them.

Media loves to project that everyone in India loves to live in clean India. However, it never talks about citizens' responsibility and the significance of doing bit by all to make India clean. Media tries to the question the responsibility of Panchayats and municipalities. Along with that, it is the media's responsibility to analyse the ground reality and the existing patriarchal attitude to address the issues.

The media can widely publish the below case study of how economic advantage made a village maintain 100 per cent sanitation.

Case Study: Mawlynnong in Meghalaya

Mawlynnong is small schedule tribe village (Khasi tribes), located in southern hills of Meghalaya under the administrative jurisdiction of Pynursula Development Block. The village is spread over 4 hectares. The main occupation of the village is a plantation. This case study helps us know how this village became the cleanest village in Asia. Cholera outbreak happened in this village centuries ago. The first batch of missionary priest convinced and sensitised the people on hygienic life to fight against the cholera epidemic. Hygiene is the only way for a healthy life. According to Rishot, a School teacher, 'their forefather taught them to live a clean, hygienic life'. Even though their forefather taught them to live clean and hygienic, they could still sustain the same till now. A few factors encouraged them to make the village clean and live clean and healthy lives. According to the people in this village Cleanliness means 'no open defecation, personal cleanliness, cleanliness in public places, and plastic-free environment'. According to the village people, "they are carrying their legacy forefathers to keep the village clean and environment friendly". This village has hundred per cent open defecation free, and this village has a designate place to rear animals. So this village is animal excreta free. As we have seen in this Unit, water is the prime concern for achieving open defecation free. The village is located in East Khasi hills. No dearth for water, still it is challenging to get piped water. They constructed check dams to store the water. People use check dams to wash clothes and utility purposes. The government provided piped water for drinking purposes. This village is a good example of social capital. They created an institutional mechanism to keep the village clean. Village level committee has been constituted under village durbar to oversee the cleanliness activities. The village committee engaged four women to sweep the village and three gardeners for beautifying the village. They made the payment to the sweepers through village council fund. They use locally available resources for village cleaning purposes. The public toilet is maintained through a user fee. They made dustbin through locally available Bamboo material and kept every place, and everyone in villages are encouraged to use dustbins. Being the cleanest village in Asia, the village could able to attract tourists. The regular arrival of tourists provided opportunities for the villagers to diversify economic activities. The annual income of villagers increased to 60 per cent with the growth of tourism. The villagers could able to demonstrate that cleanliness can also provide opportunities to grow economically.

Media must talk about data on health and sanitation, impacts of caste and culture and religion on the sanitation behaviour of the people. The media need to sensitise the people on proper health and sanitation (Ref. Pant. R.M and Mukesh Kumar Shrivastava. Cleanliness is Unique selling Proposition (USP): A case of Mawlynnong in Meghalaya).

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Write short notes on the existing health and sanitation situation in India.

.....

.....

.....

.....

.....

2) What is SQUAT Survey?

.....

.....

.....

.....

.....

5.8 LET US SUM UP

Learners must have understood the significance of health and sanitation to live a healthy life. In this Unit, we have seen the efforts taken at the international and national levels to improve the sanitation facilities. The governments of the developing countries like India have to address the sanitation issues to improve health. After reading this Unit, we must have understood the direct link between health and sanitation. The government and media need to look into the sanitation in India closely. The media must play a crucial role in bringing changes in people's behaviour towards sanitation. Along with publishing data and government programmes, media must talk culture, patriarchal attitudes of the people, and caste to address open defecation. It is high time to end open defecation in the 21st century.

5.9 GLOSSARY

Open defecation

Open defecation is defined as the practice of discharging human excreta in open fields, waterways and open trenches without any proper disposal. The term “open defecation” is used in the publications of Joint Monitoring Program (JMP) in 2008, a joint collaboration of World Health Organisation (WHO) and United Nations International Children's Emergency Fund (UNICEF) to evaluate the global progress on water and sanitation goals.

SQUAT Survey

Data collected through survey from November 2013 to March 2014. The details of the survey and questionnaire are available in the following website.

www.riceinstitute.org. sample design and methodology are also available online as squant working paper 1. The authors are also published part of working paper as Coffey, D.A Gupta ,P, 'Revealed preference for open defecation '. Economic and Political weekly, Vol.49, No.38. p.43.

DHS

Demographic and Health Survey. It is also known as National Family Health Survey (NFHS) in India.

5.10 REFERENCES AND FURTHER READINGS

Cairncross S, Hunt C, Boisson S, Bostoen K, Curtis V. 2010. Water, sanitation and hygiene for the prevention of diarrhea. *International Journal of Epidemiology*. 39:193–205

Colley DG, Bustinduy AL, Secor WE, King CH. Human schistosomiasis. *Lancet*. 2014;383:2253–64.

Dhaske, Govind and S Parasuram. 2016. Swachh Bharat Mission (SBM) and Community Approaches to Total Sanitation (CATS): Instrumental Efficacy of Integrated Micro Planning Framework. IN SivaRam,P. R Ramesh and Y Gangi Reddy (Eds). *Rural Sanitation in India Achievements, Trends and Challenges*. Hyderabad. National Institute of Rural Development and Panchayati Raj.

Pant. R.M and Mukesh Kumar Shrivastava.2016. Cleanliness is Unique selling Proposition (USP): A case of Mawlynnong in Meghalaya. IN SivaRam,P. R Ramesh and Y Gangi Reddy (Eds). *Rural Sanitation in India Achievements, Trends and Challenges*. Hyderabad. National Institute of Rural Development and Panchayati Raj.

Reddy, Sudhakar.O. 2016. Promotion of Total Sanitation in Rural Telangana through Village water and Sanitation Committees (VWSCS)- A Holistic Approach derived from State government's flagship programme "Gram Jyothi". IN SivaRam,P. R Ramesh and Y Gangi Reddy (Eds). *Rural Sanitation in India Achievements, Trends and Challenges*. Hyderabad. National Institute of Rural Development and Panchayati Raj.

Feachem RG, Bradley DJ, Garelick H, Mara DD. 1983. Sanitation and disease. Health aspects of wastewater and excreta management. Chichester: John Wiley & Sons.

Hirve S, Lele P, Sundaram N, Chavan U, Weiss M, Steinmann P, Juvekar S. 2015. Psychosocial stress associated with sanitation practices: experiences of women in a rural community in India. *Journal Water Sanitation Hygeine Development*. 5:115 –26.

Sahoo KC, Hulland KRS, Caruso BA, Swain R, Freeman MC, Panigrahi P, Dreibelbis R. 2015. Sanitation-related psychosocial stress: a grounded theory study of women across the life-course in Odisha, India. *Social Science Medicine*. 139:80 –9. 43.

SivaRam,P. R Ramesh and Y Gangi Reddy (eds). 2016. *Rural Sanitation in India Achievements, Trends and Challenges*. Hyderabad. National Institute of Rural Development and Panchayati Raj.

Unicef. 2018. Evidence Review Potential Impact of Sanitation on Health and Wellbeing Final Report. New Delhi: UNICEF

Web Sources

Health and Environmental Sanitation In India: Issues for prioritising control strategies www.ijoem.com Sunday .February 26, 2012. Ip 59.92.244. 206. Accessed on 25th December, 2020.

<https://swachhbharatmission.gov.in/sbmcms/index.htm> accessed on 25th December, 2020.

<https://www.indiawaterportal.org/articles/gender-and-safe-sanitation-rural-india> accessed on 24th December, 2020.

World Health Organisation. WHO fact sheets for schistosomiasis. Switzerland. 2017. <http://www.who.int/mediacentre/factsheets/fs115/en/>. Accessed on 24th December 2020.

World Health Organisation. WHO fact sheets for trachoma. Switzerland. 2017. <http://www.who.int/mediacentre/factsheets/fs382/en/>. Accessed on 24th December, 2020.

Coffey, Diane and Dean Spears. 2017. Where India Goes. Noida: Harper Collins Publishers India.

Dreze, J, and Amartya Sen. 2013. Uncertain Glory: India and its Contradictions. Princeton: Princeton University Press.

Spears, D and A. Thorat. 2017. Caste, Purity and Pollution and the puzzle of open defecation in India : Evidence from a novel measure in a nationally – representative survey. Economic Development and Cultural change.

Huesco, A. and B. Bell, 2013. ‘An Untold story of policy failure: The Total sanitation campaign in India’. Water Policy. Vol .15, No. 6, pp.1001-1017.

Khare, R.S, 1962. ‘Ritual Purity and pollution in relation to domestic sanitation’. *The Eastern Anthropologist*. Vol.15, No.2

Luthi, D. 2010. Private Cleanliness, Public Mess: Purity, pollution and Space in Kottar, South India Urban Pollution: Cultural Meanings, *Social Practices*, Vol 15. p.57.

Srinivas M.N, 1976. The Remembered Village. New Delhi: Oxford University Press.

5.11 ANSWER TO CHECK YOUR PROGRESS EXERCISES

1. According to the World Bank’s world development indicators, 2012 43 out of every 1000 babies born in India dies before their first birthdays. Open defecation rate is 67 percent in Gujarat and 76.8 percent in Tamil Nadu according to 2011 census.
2. SQUAT Survey tries to find out the reasons for not using toilets through different dimensions.

UNIT 6 EDUCATION AND MEDIA

Structure

- 6.0 Introduction
- 6.1 Learning Outcomes
- 6.2 Education: Concept and Role
 - 6.2.1 Literacy and Education
 - 6.2.2 Functions of Education
- 6.3 Education and Human Development
- 6.4 Education in India: A Historical Overview
 - 6.4.1 Pre-Independent Era
 - 6.4.2 Post-Independent India
 - 6.4.3 Contemporary India
- 6.5 Media and Technology for Education
- 6.6 Using ICT for Education: Some Experiences
 - 6.6.1 Initiatives in Education and Development
 - 6.6.2 School Education: Some Experiences
 - 6.6.3 Initiatives in Higher Education
- 6.7 Let Us Sum Up
- 6.8 References and Further Readings
- 6.9 Answer to Check Your Progress Exercises

6.0 INTRODUCTION

Education, as you are aware, is a vital investment in human and economic development. Education in the present day context is perhaps the most important means for individuals to build their capability levels, overcome constraints, improve competencies and in the process, enlarge their available set of opportunities and choices for a sustained improvement in their quality of life.

In this unit we shall discuss the role and importance of education, take a historical overview of education in India and analyse the role of media in education with the help of some experiences.

6.1 LEARNING OUTCOMES

After studying this Unit, you should be able to

- discuss the role and importance of education in human life;
- describe various schemes of government in the education sector; and
- analyse the role of mass media in issues related to education.

6.2 EDUCATION: CONCEPT AND ROLE

Great thinkers from Plato to Mahatma Gandhi have discussed the need and importance of education. Let us understand how great educators and thinkers of

the east and west have defined the role of education. According to the Greek philosopher Plato, “Education develops in the body and the soul of the pupil all the beauty and all the perfection he is capable of.” It is an action upon our mental nature and is the remuneration of years of labour. Aristotle mentioned, “Education is the creation of sound mind in a sound body. It develops man’s faculty, specially his mind, so that he may be able to enjoy the contemplation of supreme truth, goodness and beauty.” While Rousseau, a political philosopher, stated, “Education is the child’s development from within”. Swami Vivekananda affirmed, “Education as the manifestation of perfection already in man. Like fire in a piece of flint, knowledge exists in mind. The suggestion is the friction, which brings it out.” Mahatma Gandhi observed, “By education, I mean an all-around drawing out of the best in the child and man’s body, mind and spirit.”

From the above definitions, it is apparent that education is a broad concept. From ancient times to today’s modern societies, education has played a crucial role in the holistic development of individuals by enabling learning by gaining knowledge and the development of skills, nurturance of values and beliefs, and the physiological and psychological growth of individuals.

6.2.1 Literacy and Education

The terms Literacy and education are often interchangeably used; however, they are not identical concepts. Literacy is the ability to read and write. When you can read a book or a newspaper, write a letter or do mathematical calculations, these are all examples of literacy. People differ in their literacy skills, which is a gradual progression, beginning with the ability to understand spoken words, decode written words, and deeper understand the meaning in texts. On the other hand, education is a much more significant concept that leads to the development of the capabilities of individuals and enhances their competence and skills to meet the challenges of life. Literacy helps acquire skills and learning, while education is about applying these skills and learning to analyse situations comprehensively. Hence every literate person cannot be called an educated person and vice versa. Education is a continuous lifelong learning process where everyday activities and events that we experience educate us in one or the other. Improvement in education level indicates improvement in the quality of the human resource.

6.2.2 Functions of Education

The constitution of India resolves to secure justice, liberty and equality for all its citizens. As society develops, it becomes imperative that the cumulative experience and the knowledge necessary for social, economic, and political development should be passed on to new generations and people who need this knowledge. To realise these noble goals, education plays a significant role in any society performing several crucial functions.

- Education provides the value of unity which leads to integration in society. Education expands our outlook and teaches us to be tolerant towards other views. Educational institutions provide new skills and opportunities to interact with people of different social milieu, making an educated person receptive to various points of view. Education broadens our intellectual panorama, leading to greater enlightenment.
- Awareness is an asset in itself, and education broadens our awareness and makes us knowledgeable about a range of issues and our privileges and

duties. An aware society leads to positive behaviour and the growth of humankind.

- Education enhances citizen's participation in democratic processes so that they become responsible citizens. It enables us to understand our rights as well as responsibilities and encourages us to follow them. An educated person is a more conscientious citizen who sensibly exercises adult franchise and selects able leaders.
- Education leads to awareness which facilitates decision making. Well informed decisions enable us to make the right choices, prevent losses and thus contribute to the per capita income and the economy of a country. To realise the Sustainable Development Goals and the overall development of nations, education is a significant factor.
- Education prepares an individual for economically productive activities and the workforce. Our productivity increases manifold by acquiring new skills and competencies and enabling individuals to augment and earn a continual livelihood. Education is futuristic; one who receives education secures their future.
- Education is significant to secure justice, liberty and equality for all citizens. It helps in preparing an individual to work in the economy, teach values and morals. The impact of Education enables people to lead healthy lives. A well-educated individual would be physically and mentally strong, thus having better opportunities in life.

6.3 EDUCATION AND HUMAN DEVELOPMENT

Education is considered an essential component/factor for eradicating poverty and sustenance of fair democratic processes in today's societies. Millions of people worldwide are unable to play their entire part in their communities and nations' social and economic life because they lack the basic skills to read or write. Literacy is considered vital for the overall development of individuals, boosts economic growth, enhances people's skills and livelihoods, and improves people's lives and a nation's overall development. Sustainable Development Goal (SDG) 4 of ensuring inclusive and equitable quality education and promoting lifelong learning opportunities is essential for achieving sustainable development.

The concept of Human Development (HD), as it was formulated in the first Human Development Report (HDR) in 1990, identifies education as a critical component in advancing HD. Education is one of the three dimensions of the index. HDI is a composite index of assessing progress in three basic dimensions of human development — health as measured by life expectancy at birth, knowledge measured by mean years of education, and standard of living measured by per capita gross national income. HDRs have annually explored critical development themes through the HD lens and have reinforced the importance of investing in education.

India's Human Development Index (HDI) value for 2019 is 0.647, which puts the country in the medium human development category. Between 1990 and 2019, India's HDI value rose from 0.427 to 0.647. India climbed up a spot to rank 129 out of 189 countries in the latest human development rankings by the

United Nations Development Programme (UNDP). According to the Census of 2011, “every person above the age of 7 years who can read and write in any language is said to be literate”. According to this criterion, the 2011 survey holds the National Literacy Rate to be around 74.07%. Government statistics of 2001 also hold that the increase in literacy rate is more in rural areas than in urban areas. Within the Indian states, Kerala has shown the highest literacy rates of 93%, whereas Bihar averaged 63.8% literacy. An effective education system is fundamental for a developing nation like India. It is of utmost importance to understand our present education system by incorporating sustainable changes to make it compatible with the global standards.

6.4 EDUCATION IN INDIA: A HISTORICAL OVERVIEW

Education in India has always been valued more than merely considering it as a means towards earning a good living. Right from pre-historic days, education, especially higher education has been given a predominant position in the Indian society. Great universities flourished in India. German scholar Max Muller observed: “If I were asked under what sky the human mind has most fully developed some of its choicest gifts, has most deeply pondered on the greatest problems of life, and has found solutions to some of them which well deserve the attention of even those who have studied Plato and Kant, I should point to India”. When we are studying a unit on Education and Media it is important to have a brief glimpse of the historical overview of Education in:

- Pre- independent era
- Post-independent era
- Contemporary India

6.4.1 Pre-Independent Era

Education System prevalent in India during the ancient vedic period was based on the ‘Gurukul’ system, where a student or ‘shishya’ had to stay away from his home at the house of a teacher/‘Guru’ so as to develop a close bond between the student and the teacher. While living in a gurukul, the shishyas would be disconnected completely from their families, for long periods of time. In the gurukul system all students were treated as equals, irrespective of their social background. It worked towards the holistic development of students by not only making them socially aware, but also building their personalities and character, imbibing self-control and discipline and nurturing their intellectual, spiritual and physical development. The relationship between a guru and the shishya was considered very sacred and at the end of one’s education, a student offered the guru dakshina before leaving the gurukul which was an acknowledgment of respect and gratitude to the guru.

In ancient India, education was given immense importance and was considered as an instrument which puts an ignorant person on the path of intellectual progressive, moral and virtuous course of life. Later, powerful kings took keen interest in promoting the interests of higher education by giving rich donations and land to learned scholars. The major universities – Nalanda and Takshila in ancient India were the seats of great intellectual scholarship.

British colonial rule brought with it the concept of a modern state, a modern economy and a modern education system. In British period Western education was introduced in India. This period was of great historical significance for emergence of modern education systems in India. The British aggressively worked to spread education in India, using it as a tool for furthering their hidden motives among the natives of India along with education.

In the pre-independence period, the British government was the main agency for deciding the direction and focus of the education system in India. They facilitated in launching a number of schools and colleges throughout the country from where several educated Indians, well versed in modern subjects graduated. British government's main purpose was not to produce thinking intellectual minds but to produce clerks for their administrative machinery. During this time social problems increased manifold and several scholars felt the need for introduction of English education in India.

However, this new western culture and education gave birth to a new awakening as well. Several western educated social thinkers, contributed in rousing the national consciousness in the minds of Indians in the nineteenth and the twentieth centuries. The main torch-bearer of this new found awakening was Rajaram Mohan Roy, the founder of Brahmo Samaj. Other social and religious reformers included Swami Dayanad Saraswati who founded the Arya Samaj, the Ramakrishna Mission started under the leadership of Swami Vivekananda and the Theosophical Society of Annie Besant. Along with other social reformers such as Ishwar Chandra Vidyasagar, Sri Aurobindo Ghosh they initiated novel ideas of a new type of education for the country. These movements, while trying to reform religion and society, were fully aware of the importance of education as a tool in arousing the national consciousness and national regeneration. Their leadership and legacies substantially helped in evolving a national system of education in India.

6.4.2 Post-Independent India

After independence, India adopted the Constitution in 1950 and education became the responsibility of both – the state and the central governments. Following Independence, school curricula were thus imbued with the twin themes of inclusiveness and national pride, placing emphasis on the fact that India's different communities could live peacefully side by side as one nation. Subsidised quality higher education through institutions such as the Indian Institute of Technology (IITs) and Indian Institutes of Management (IIMs) formed a major contribution of a self-reliant and modern Indian state, and they now rank amongst one of the premier higher education institutions in the world.

The independent India resolved to secure justice, liberty and equality for all citizens to achieve these noble goals education. As society develops, it becomes imperative that the cumulative experience and the knowledge necessary for political, economic, social and other development should be passed on to new generations. To this effect, the Indian Constitution under its various articles mandates that free and compulsory education be provided as a fundamental right to children between the ages of 6 and 14.

India has made substantial progress in increasing the achievement of universal primary education. Since independence, Almost 75% of the population, aged between 7 to 10 years, was literate in 2011. Enrollment in higher education has also increased steadily reaching a Gross Enrollment Ratio of 26.3. However Indian education still lags on many fronts compared to developed nations and if India has to reap the benefits of its demographic dividend from its young population its education sector needs to make tremendous strides.

The challenges before the Indian education system have been succinctly summed up in NGO Pratham report which states that, “Modern education in India is criticised for encouraging rote learning, rather than comprehension, critical thinking, and problem solving. Students spend most of their time memorising a syllabus with no thought given to learning or playing. Textbook knowledge, rigid ideas, and test scores take precedence over open debates and logical reasoning. Little room is left for creativity to thrive. Moreover, there are growing concerns about student learning outcomes, teacher training, curriculum quality, assessment of learning achievements, and the efficacy of school management. Faced with such problems, many children drop out of school before even completing five years of primary education. Those children who do stay on often learn little”.

Juxtaposed with these challenges is the new technological revolution, the effects of which, all segments of society are experiencing. Today’s students are the first generation to grow up with digital technology which has changed the way they view knowledge, access information and relate to our world. Now, technology is offering new possibilities and making life easier for both students and educators.

Information and Communication Technology (ICT) is basically an umbrella term that encompasses all communication technologies such as, internet, wireless networks, cell phones, satellite communications, digital television, radio, etc. Educational institutions are increasingly adopting digital teaching-learning solutions. Education needs to both capture the incredible possibilities for deep learning opportunities that new technologies can offer and prepare students to cope with the amount and speed of information at their fingertips.

Online education in India has been gaining importance in the recent times. Online facilities have tried to address the challenges of delivering education in remote areas. Educational technology can enable more individualised instruction and student teacher interaction that allow for better differentiation enabling students to learn at their own pace. Online education has become an effective mode of learning and the growing awareness of various online courses is encouraging more students to opt for this mode of learning. People choose online methods of education for various reasons; some do it to acquire skills independently while others do it to gain accreditation from institutions worldwide in order to gain better qualifications as per their convenience without withdrawing from their work. Online learning is a cost effective and faster mode of learning.

A large number of distance education universities and programmes use ICT to supplement and complement the self-learning materials that they deliver to students. These include audio-video broadcasts such as radio and television programmes, delivered to students as part of a learning kit, and in more recent times, multimedia content such as lessons which are delivered off line in CDs.

This is also sometimes called multimedia education, where multiple media are used to support learning.

Multimedia education is an interactive instructional method that uses a computer to present material, track learning and direct the user to additional material which meets the students needs. Multimedia learning uses a combination of text, graphics, sound and video in the learning process. It can also be used to describe Internet based instruction through the use of Web Pages, Web Bulletin Boards, Letters and Newsgroups, Video and Real Audio, Graphics and Hands-on Applications.

In India, we have a 24-hour education channel- Gyan Darshan-as part of the national repertoire and GyanVani - the radio co-operative to remove the remoteness through the use of communication technologies. The responsibility of running them has been essentially entrusted to the Indira Gandhi National Open University (IGNOU), but more collaboration and networking is needed in this regard.

The concept of e-learning in India is in its early stages and as a mode of teaching and learning, it has seen an increasing use in educational institutions in cities.

Activity 1

Search the internet for Educational Technologies that are being used in India. What are the advantages and limitations of using these technologies?

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is the difference between Education and Literacy?

.....

.....

.....

.....

.....

.....

2) What is ICT?

.....

.....

.....

.....

.....

.....

6.5 MEDIA AND TECHNOLOGY FOR EDUCATION

According to Rowntree, communication modes and media are essential for the effective transfer of learning as they enable the student to use different senses to comprehend concepts. Romiszowski reiterates that the concept of communication is closely linked to learning. Education hence is essentially a communication process. The communicator and the receiver (the communicator may be teacher or student, and the receiver may be student or teacher here) partake in a two-way exchange for the student's growth, development, and learning. The evolution of educational technology has seen shifts in its approach and focus. Lewis Elton (1977) identifies three broad transfers: mass communication, individualised learning, and group learning.

In the earlier phase, the focus was more on technologies like Radio and TV, with more importance given to technology to achieve cost-effectiveness and reach out to many students. Later shifts towards individual learning and group learning led to more emphasis on individual need, constant feedback and peer support in the learning process. The role of educational technology shifted from playing an instructional to a more supportive role for enhancing the quality of education.

For solving the problem of “more education to more people in less time”, educational media and technology have a crucial role as they can tremendously influence the structures, processes and outputs of teaching-learning. The ever increasing presence of media in the Indian subcontinent makes it a potential instrument of mobilising public opinion. If positively harnessed, it can play a crucial role in motivating parents to send their children to school and reduce dropouts. Media can help to deliver the learning materials from communicator to receiver as per the objectives of communication. With the advent of internet the pedagogy of education has seen a sea change. Internet has been used to expand access, promote efficiency, improve quality of learning, enhance the quality of teaching, and improve management systems.

The experience of introducing different media and communication technologies in the classroom and other educational settings worldwide over the past several decades suggests that the full realisation of the potential educational benefits of ICTs is not automatic. The effective integration of media technology into the educational system is a complex, multifaceted process that involves curriculum and pedagogy, institutional readiness, teacher competencies, and long-term financing.

Media Technologies are potent tools for extending educational opportunities for various sections of the population. It also provides opportunities for formal and non-formal education. Media Technologies may enhance the opportunities for the rural population, women, girls, marginalised, socially and economically excluded, ethnic minorities, persons with disabilities, elderly, and those who are unable to enrol in on-campus programmes due to cost or time constraints.

- **Flexibility, Adaptability:** A feature of media technologies is their flexibility for various tasks for both educators and students. Educational media technologies can be useful tools for how we structure tasks to better understand procedures and processes; access knowledge databases and find

information that is appropriate and relevant; as well as provide us with multiple forms of knowledge representation (audio, text, video, image etc.) styles that we can choose from. Thus ICTs can effectively offset variations in student-teacher abilities in the learning process.

- **Anytime, Anywhere:** One defining feature of Media and Communication Technologies is their ability to transcend time and space. With access to the Internet, distance educators and their students can use email and Web search engines to get more information and share the information. As the Internet continues its rapid penetration into households, education in virtual universities and distance education continues to develop. Distance education serves to reverse social dynamics by bringing the school to students rather than students to school. Media and Communication Technologies make possible asynchronous learning or learning characterised by a time lag between the delivery of instruction and its reception by learners. Online course materials, for example, maybe accessed 24×7. ICT-based educational delivery (e.g., educational programming broadcast over radio or television) also dispenses with the need for all learners and the instructor to be in one physical location. Additionally, specific ICTs, such as teleconferencing, enable instruction to be received simultaneously by multiple, geographically dispersed learners (i.e., synchronous learning).
- **Access to remote learning resources.** Information technology has made education accessible in remote areas allowing teaching and learning to take place beyond the traditional boundaries and resources of the school. Teachers and learners no longer have to rely solely on printed books and other materials in physical media housed in libraries (and available in limited quantities) for their educational needs. With the Internet and the World Wide Web, a wealth of learning materials in almost every subject and various media can now be accessed from anywhere and anytime by an unlimited number of people. Students in remote and under-developed areas are the largest beneficiaries of education through the Internet. Students who cannot get through university entrance examinations and working people can also get a chance of continuous education and training at virtual universities. This is particularly significant for many schools in developing countries and even some in developed countries with limited and outdated library resources. ICTs also facilitate access to resource persons - mentors, experts, researchers, professionals, business leaders, and peers worldwide.

The very origin of the Internet is strongly linked to education, as it was in universities and research institutes that electronic networks were initially developed. Internet technology leads to the growth of distance learning, which gives millions of people who lack the time or resources to attend traditional schools and colleges to pursue education qualifications at their own pace. Because of distance learning, students who live in remote areas no longer have to travel great distances to get to classes or access educational materials. While this is particularly true in rural and remote areas, it applies to many urban areas.

There are three general approaches to the instructional use of computers and the Internet, namely:

- Learning about computers and the Internet, in which technological literacy is the end goal;

- Learning with computers and the Internet, in which the technology facilitates learning across the curriculum; and
- Learning through computers and the Internet, integrating technological skills development with curriculum applications.

As computer and internet-based teaching tools become popular innovations in education, it is vital to keep in mind the desired results before introducing educational technology. How is it going to bring an increase in the quality of learning? Will there be a decrease in the time taken for the learner to attain the desired goal? Is there an increase in the capacity of teachers in terms of learners taught without reducing the quality of learning? Is educational technology helping to reduce cost in the long run without affecting quality? A holistic understanding of media and technologies can go a long way in strengthening teaching learning problems encountered in educational institutions.

6.6 USING ICT FOR EDUCATION: SOME EXPERIENCES

In the previous section we have discussed that ICTs which include radio and television, as well as newer digital technologies such as computers and the Internet are potentially powerful enabling tools for educational change and reform. When used appropriately, different ICTs help expand access to education, strengthen the relevance of education to the increasingly digital workplace, and raise educational quality. ICTs also make teaching and learning into an engaging, active process connected to real life. Some of the initiatives in India using ICT for education are as follows:

6.6.1 Initiatives in Education and Development

In India, media technologies have been used to facilitate education and development through projects such as SITE, Kheda and JDCP.

SITE: India was one of the first developing countries to experiment with satellite television. The Satellite Instructional Television Experiment (SITE) was a technology based experiment conducted in India in 1975-76. It was designed in collaboration with NASA and the Indian Space Research Organisation (ISRO). The Satellite SITE was the first attempt to use technology as an educational tool. In 1975, when television was a rare even in urban India, TV sets for community viewing were set up in schools or Panchayat centres in 2,400 villages in six States — Bihar, Orissa, Madhya Pradesh, Andhra Pradesh, Karnataka and Rajasthan. The programmes were directed at creating a positive attitude to formal education and making education interesting, creative, purposive and stimulating.

Kheda Communication Project: Kheda Communication Project was yet another pioneering projects using television for educational purposes in Kheda district of Gujarat, India. In this venture villagers were encouraged to participate as actors, writers, in the production of T.V. programmes which was based on the community issues such as alcoholism, family planning etc. Chatter Mota and NariTu Narayani were two popular soaps of this project. This initiative aided in facilitating services based on the needs of community, community mobilisation and acted as a medium to initiate a dialogue among locals. This project

demonstrated how technology can be put to use for social progress and development especially for ones who are in the deprived segments of society.

Jhabua Development Communication Project: Jhabua Development Communication Project (JDCP) was launched in mid 1990s. It used satellite communication to address the needs of the rural illiterate population and provide programme support communication to development efforts. The project was located in Jhabua, a predominantly rural area with a large tribal population in the state of Madhya Pradesh in central India. The purpose of JDCP was to experiment with the utilisation of an interactive satellite-based broadcasting network to support development and education in remote areas of India. Some 150 direct-reception systems like satellite dish, TV sets, VCRs, and other equipment have been installed in several villages of Jhabua, which received television broadcasts for two hours every evening from Ahmedabad studio, uplinked through satellite. Moreover, 12 talkback terminals have been installed in each of the block headquarters of Jhabua district, through which village functionaries ask questions, provide feedback, and report on the progress.

6.6.2 School Education: Some Experiences

Several private and government projects have harnessed ICT for education at the school level, some of these include:

Hole in the Wall Project: An experiment for children's learning - "Hole in the Wall" (HIW) was first conducted in 1999 in which a computer was placed in a kiosk in a wall in a slum and children were allowed to use it freely. The experiment was conducted with a purpose to demonstrate how technology can be used to teach children without formal training. Through this experiments, especially with children from resource poor settings it was concluded that the key is forcing a group of children to share a computer rather than giving one to each, since sharing foments debates and a collaborative knowledge-seeking process which is far more effective than learning alone. This showcases the potential of ICTs to bring about educational change and how these can be effectively used with children.

Gyandoot Samiti: The Gyandoot Samiti, established 32 kiosks at the high schools and higher secondary schools to provides them local educational contents through the server. The schools are also accessing the Internet for other educational contents. Each school is having an e-club to promote activities related to IT among the rural students. This initiative of the Samiti has been named Shiksha Gyandoot. It is meant for the benefit of both students and teachers to foster the motto of providing good quality education to all. At present the Samiti provides the following facilities to the students through its site:

- **Career guidance:** students of various streams (PCM, PCB, Commerce, Arts etc.) with the various options open to him after completion of his XII board.
- **Pathyakram:** syllabuses of various subjects taught in class X and XII. and question bank on various subjects available to students to help them prepare better for the exams
- **Chhithi:** This option allows the student to interact with other schools, soochnalayas etc. and discuss their problems and experiences.

- **General Awareness:** A question bank, which allows the student to assess his/her general awareness.
- **Prerak Prasang:** Moral stories to help the students develop good moral values.
- **Jiwaniya:** Biographies of great leaders in various fields of our country. Presently it includes the biographies of all the Prime ministers and Presidents of our country besides scientists and mathematicians like Arya bhatt, Vikram Sarabhai etc.
- **Sawaliram se poochiye:** opportunity provided to schoolchildren to ask inquisitive questions regarding career counseling or any other field from Sawaliram at no cost.
- **Vidya Vahini:** In year 2003, the Government launched “VidyaVahini”; an ambitious school computerisation programme aimed at connecting 60,000 Government aided schools through internet and intranet. The Schools were given computer labs to facilitate IT education, access to Internet, online library, academic services, web broadcast and e-learning.
- **ePathshala:** In the year 2015, the Ministry of HRD and NCERT jointly launched an educational portal called ePathshala. Available in three languages – English, Hindi and Urdu – it provides educational resources for students, teachers, as well as parents. A web and mobile based application, it features textbooks, audio visuals, print material, etc.

6.6.3 Initiatives in Higher Education

ICT have been used to facilitate higher education in the country. Some such initiatives have been discussed in this section.

- **Country Wide Class Room Project:** The ‘University Grants Commission’ (UGC) in collaboration with ‘INSAT’ launched educational television project, popularly known as ‘Country wide Classroom’. The project was launched on August 15, 1984. It was the first project which targeted the under-graduates in India. The media centres were established in 6 universities of India. These centres were named Audio Visual Research Centres (AVRCs) (AVRCS) (later these centres have been renamed as Educational Multimedia Research Centres (EMMRCS). The project used electronic media for the quality enrichment of higher education.
- **UGC-INFONET Programme:** Indian Universities constitute one of the largest higher education systems in the world. With large number of universities and affiliated colleges it is a great challenge to ensure effective coordination and communication amongst students and teachers. Frequent changes in curricula and introduction of new subjects impose a great demand on the system in general. Indian Universities need to be given the required trust to enter the third millennium with a leading edge.

Technology is a driving force in contemporary education system. University Grants Commission has launched an ambitious programme to bring about a qualitative change in the academic infrastructure especially for higher education. Under this initiative UGC is modernising the university campuses with state-of-the-art campus wide networks and setting up its own nationwide communication network named UGC-InfoNet.

ERNET India, a society under the Ministry of Communications and Information Technology, in partnership with the University Grants Commission has set up UGC-InfoNet. Under this programme it is proposed to use ICT and Internet to transform learning environment from mono-dimensional to multi-dimensional. UGC-InfoNet will be a boon to the higher education system in many ways:

- A vehicle for distance learning to facilitate spread of quality education all over the country.
- A tool to distribute education material and journals to remotest of areas.
- A resource for researchers and scholars for tapping most up-to-date information.
- A medium for collaboration among the teachers and students not only within the country but all over the world.
- Will be an Intranet for University Automation.
- Will encompass entire University System for most efficient utilisation of precious network resources.
- Will establish a channel for Globalisation of Education and facilitate the universities in marketing their services and development.

Gyan Darshan

Gyan Darshan channel is a major milestone in the field of Educational Television in India. It is a joint venture of the Ministry of Human Resource Development (MHRD), Ministry of Information & Broadcasting (I & B Ministry), Prasar Bharati and IGNOU serving as the nodal agency. Launched in the year 2000, Gyan Darshan is a 24-hour educational channel which offers educational programmes covering a variety of subjects and catering to a wide range of viewers. These include pre-school, primary, secondary and higher secondary students, college/university students, youth seeking career opportunities, homemakers and working professionals. The software is pooled from various educational Institutions and Development Organisations. GD conducts two hours of live interactive sessions every day to build interactivity in the Open and Distance Learning (ODL) system. The Gyan Darshan telecast is also beneficial for students of the formal education system and the viewers can access Gyan Darshan on IGNOU's website <https://www.ignouonline.ac.in/gyandarshan/>.

Gyan Vani

Gyan Vani (GV) FM Radio was conceived in 2001 as a network of educational FM Radio Channels operating from various cities in the country. With an aim to enhance and supplement the teaching-learning process, each GV Station has a range of about 60 kms and covers an entire city/town including the adjoining rural areas. Gyan Vani serves as an ideal medium for niche audience addressing the local educational, developmental and socio-cultural requirements of the people. The flavour of the channel is by and large local and the medium is English, Hindi or language of the region. The overall content pertains to Primary and Secondary Education, Adult Education, Technical and Vocational Education, Higher Education, Distance Education and Extension Education etc. Interactive Radio Counseling (IRC) facility is being provided by GV Stations to enable students to interact with the faculty, academic counselors and student support staff. The live phone-in programmes are popular components of the network. Students can listen to these live discussions by the teachers and experts on the

topic of the day and interact with them through telephone, email or through chat mode on Gyan Dhara.

Gyandhara

Gyandhara is an internet audio counseling service offered by IGNOU. Students can listen to the live discussions by the teachers and experts on the topic of the day and interact with them through telephone, email and also chat mode. The Gyandhara streaming is available for internet users anywhere in the world. Important events broadcast by GV Delhi are also relayed by all GV stations using the Gyandhara feed. You can access Gyandhara using the link <https://www.ignouonline.ac.in/gyandhara/>

eGyanKosh

eGyanKosh is a National Digital Repository to store, index, preserve, distribute and share the digital learning resources developed by the Open and Distance Learning Institutions in the country. Items in eGyanKosh are protected by copyright, with all rights reserved by IGNOU, unless otherwise indicated.

Swayam Prabha

IGNOU is the national coordinator for five channels of Swayam Prabha, the DTH channel initiative of Government of India. IGNOU has been coordinating with INFLIBNET, MHRD and for production of videos, scheduling and management of these channels.

From above discussion you would have understood that media and ICT play an important role in improving the Indian education system in India.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is SITE?

.....

.....

.....

.....

2) Briefly outline the role of internet in the progress of Indian education system.

.....

.....

.....

.....

6.7 LET US SUM UP

In this Unit you were exposed to various aspects of media and education. We started with the concept and functions of education and made a distinction between literacy and education. We discussed the significance of education and examined

how education contributes in our overall development and plays an important role in democracy too. Then we looked at the historical overview of education system in India that prevailed in the pre-independent and post-independent India. We outlined some issues pertaining to the education system in India and looked at some initiatives of using media and ICTs at the school and higher education levels. The analysis will help you in understanding the education system and harnessing media for education and development.

6.8 REFERENCES AND FURTHER READINGS

Cuban Larry. 1986. 'Teachers and Machines: The Classroom Use of Technology Since 1920'

Abbott, C. 2001. "ICT: Changing Education", Routledge/Falmer

Raman, S.A. 2006. "Women's Education", Encyclopedia of India (vol. 4), edited by Stanley Wolpert, 235–239, Thomson Gale: ISBN 0-684-31353-7

Online resources

<https://www.pratham.org/wp-content/uploads/2020/11/aserreport2019earlyyearsfinal.pdf>

<https://www.pratham.org/wp-content/uploads/2020/11/aser2020fullreport.pdf>

<http://hdr.undp.org/en/global-reports>

<https://en.unesco.org/pressroom>

<https://www.youtube.com/watch?v=e4-sIRzkoFM>

<https://www.youtube.com/watch?v=aVWHp8FsV1w>

https://www.youtube.com/watch?v=l0s_M6xKxNc

6.9 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

1. Literacy is an ability to read and write while education aims to make the human beings capable and develops their competence and skills to meet the challenges of life.
2. ICT is an umbrella term that encompasses all communication technologies such as, internet, wireless networks, cell phones, satellite communications, digital television, radio, etc. that provide access to information.

Check Your Progress Exercise 2

1. Satellite Instructional Television Experiment (SITE) was an experiment conducted in India during 1975-76 using satellite based communication. The programmes were directed at creating positive attitudes to formal education and making education interesting, purposive and stimulating.
2. Internet has facilitated in accessibility to education in far flung, remote areas. It has made teaching and learning an engaging process and raised the quality of education.

UNIT 7 MEDIA AND ENVIRONMENT

Structure

- 7.0 Introduction
- 7.1 Learning Outcomes
- 7.2 People, Planet and Sustainability
- 7.3 Environmental Risks
- 7.4 Mass Media and Environment
 - 7.4.1 Environmental Communication
 - 7.4.2 Environmental Journalism
 - 7.4.3 Functions of Mass Media towards Environment
 - 7.4.4 Media Platforms and Environmental Awareness
- 7.5 Environmental Movements
 - 7.5.1 Environment Conservation Movements in India
 - 7.5.2 “Save our Tigers” Campaign
 - 7.5.3 Media Advocacy in Environmental Organisations
- 7.6 International Environmental Agreements
 - 7.6.1 Key International Conventions and Protocols
 - 7.6.2 U.S. India Partnership on Climate Change
- 7.7 Let Us Sum Up
- 7.8 References and Further Readings
- 7.9 Answer to Check Your Progress Exercises

7.0 INTRODUCTION

Development is inevitable in a progressive society. However, when development happens relentlessly and does not account for the adversities it causes to the environment, it poses a threat to the survival of humankind. Human beings are a part of nature; they are increasingly exploiting it for selfish reasons. Humans and nature form an inseparable part of life support system which has five elements- i.e. air, water, land, flora and fauna which are inter-related and interdependent. Deterioration in any one element affects the remaining four. A short-term deterioration may repair itself, but if it continues for a long term it has the capacity to throw the entire system off balance. It is imperative and our moral responsibility to keep Earth safe for future generations.

Environmental issues such as global warming, pollution and wildlife conservation have been a concern for the scientific community for decades. However, it is only recently that public is increasingly becoming aware of such issues. This can be partially attributed to the way mainstream/news and other media have portrayed these issues alongside other significant events of the world.

Since 1990s, the shift from traditional to new media has indicated substantial changes in patterns of accessibility and interactions of people with information i.e. who has access to it, and who are considered “authorised definers” of various dimensions of environmental issues. The news serves to identify issues that are important, works as a platform for advocacy and change, and provides information about environmental crises for civic participation and autonomy.

In this unit we shall try to understand the relationship of media and environment and why it is so important in the present context. This unit highlights the role played by media and communication processes in the public and political definition, elaboration and debates over environmental issues and problems.

7.1 LEARNING OUTCOMES

After Studying this Unit, you should be able to:

- discuss the relationship of environment and human beings;
- describe some environmental challenges;
- analyse the responsibility of media towards environment; and
- examine some environmental campaigns in India and the role of media in those campaigns in India.

7.2 PEOPLE, PLANET AND SUSTAINABILITY

Nature provides raw materials for industries, food for people, fuel for transport, etc. Environment absorbs waste that developmental activities generate. It acts as both - a source and sink for human activities. However, the ecological imbalance ensued from depletion of earth's resources leads to concerns about its life support systems. For achieving sustainable development, the well-being of humans and the ecosystem are equally important and a sustainable society needs to achieve both together. In 1997, United Nations' definition of sustainability "Development is a multidimensional undertaking to achieve a higher quality of life for all people. Economic development, social development and environmental protection are interdependent and mutually reinforcing components of sustainable development", was adopted in its Agenda for Development.

To understand the co-existence of humankind and environment, International Union for Conservation of Nature (IUCN), the World conservation union, developed the "Egg of Sustainability" model. It comprises of people (human, communities, economies) within the ecosystem (ecological communities), processes and resources together with their interactions. These interactions consist of bi-directional flows – from the ecosystem to people which includes benefits (life support, economies, resources) and stresses (natural disasters) and conversely from people to the ecosystem, in terms of both the stresses (resource depletion, pollution) and benefits (conservation). Being a subsystem of the ecosystem, people depend on it; it surrounds and supports them just as the white of an egg surrounds and supports the yolk. At the same time, a healthy ecosystem is no compensation if people are victims of poverty, misery, violence or oppression. Just as an egg can be good only if both the yolk and the white are good, similarly a society can be well and sustainable only if both the people and the ecosystem are in a state of equilibrium. For a sustainable society we need to achieve this balance with effective harmony, and people need to understand that policy decisions, especially decisions on environment related issues have profound influence on their lives as well as the future generations.

In 2015, the United Nations and the international community adopted a new global agenda – Sustainable Development Goals. Health is centrally positioned within the SDGs in SDG no. 3 – Ensure healthy lives and promotes wellbeing

for all ages, but its accomplishment means more than just enhancing healthcare, it also incorporates environmental health. Thus, environment underlies each of those 17 goals, from eliminating hunger to reducing inequalities to building sustainable communities around the world. However, these SDGs directly indicate the urgency to address environmental risks – SDG 6: Clean Water and Sanitation; SDG 7: Achieving Affordable and Clean Energy; SDG 11: Sustainable Cities and Communities; SDG 12: Responsible Consumption and Production; SDG 13: Climate Action; SDG 14: Life below Water; and SDG 15: Life on Land.

7.3 ENVIRONMENTAL RISKS

In this section, we shall discuss various environmental challenges that the world faces.

Air pollution and transportation: Air pollution kills more than 6 million people every year; it is the biggest environmental health risk of our time. Air pollution also causes alterations to earth's climate, with profound effects on the health of humankind. Air pollution comes from many sources – from cooking stoves and kerosene lamps to coal-fired power plants, vehicle emissions, industrial furnaces, wildfires, and sand and dust storms. The problem escalates in urban areas, particularly in Africa and Asia. In low- and middle-income countries, 98 per cent of cities with more than 100,000 inhabitants fail to meet the World Health Organisation's air quality guidelines.

As economies grow, escalated transport activity around the world, especially in urban cities, implicitly suggests rising emissions from the sector. 95 per cent of the world's transport energy is obtained from fossil fuels. This poses a threat to environment and human health.

Energy: Energy production and use is the single biggest contributor to global warming. This sector accounts for about two thirds of global greenhouse gas (human-attributed) emissions. More than a billion people do not have access to electricity, while 3 billion use dirty fuels – charcoal and animal waste – for cooking and heating. The challenge is to reduce reliance on fossil fuels to produce electricity, while making reliable, clean and affordable energy available to everyone.

Technology: Technologies used for treating waste help solve environmental problems, while air conditioning and refrigeration systems use ozone- depleting refrigerants, which have huge environmental repercussions. How fast human society meets these and other challenges depends in large part on the pace and scale at which good technology displaces inferior technology in different global contexts.

Chemicals and waste: It forms an integral component in everyday life, but they also have major impacts on environment and human health. As the world's population nears 8 billion, there is an urgency to promote sound management of chemicals and waste. The United Nations estimate productions of 2.2 billion tonnes of waste by 2025.

Ecosystem degradation: Humans depend on ecosystems to meet their basic needs, but it is crucial that their needs be met sustainably. Biodiversity loss and ecosystem degradation are expected to continue, or even accelerate. By 2030,

the world will require 40 per cent more water, 50 per cent more food, 40 per cent more energy and 40 per cent more timber and fiber. The only way to meet these demands is by managing ecosystems smartly and sustainably, so as to make it healthy and productive (Integrated Ecosystem Management).

Water scarcity and pollution: Oceans regulate the climate, and generate most of the oxygen. They also support sectors from tourism to fisheries to international shipping. Nonetheless, oceans are facing unprecedented threats as a result of human activity. Approximately 8 million tonnes of plastic waste end up in the oceans every year. Climate change damages coral reefs and other water ecosystems. Further, overfishing threatens the stability of fish stocks; nutrient pollution contributes to the creation of dead zones; and nearly 80 per cent of the world's wastewater is discharged without treatment.

Demand for water – for drinking, sanitation, farming and energy production, among many other uses – grows with the rise in population. At the same time, human activity and climate change disrupt the natural water cycles, acting as a stressor for freshwater ecosystems. Pollution, infrastructure development and resource extraction pose additional challenges.

Climate change: It is one of the most pervasive and threatening environmental issues with far reaching impacts on the entire planet. Climate change is expected to have unprecedented implications on where people can settle, grow food, build cities, and rely on functioning ecosystems for the services they provide. In many places, temperature changes and sea- level rise are already leading to ecological imbalance and affecting human wellbeing.

Disasters and conflicts: In the last two decades, the world has been a victim to more than 2,500 disasters and 40 major conflicts. These tragic events have affected more than two billion people. Such environmental hazards destroy infrastructure, displace populations, and hamper human security. They also compound poverty and challenge sustainable development.

Gender differences: Men and women relate to environment in different ways, and environmental changes have different adversities in their lives. Women play a critical role in sustaining communities and managing natural resources, but their contributions are often undervalued and neglected. Women are more likely than men to live in poverty, and they are more vulnerable to the impacts of climate change, especially in developing countries. These differences are often magnified by other social determinants such as age, socio-economic status, and geographic location.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Enlist any three ways that transports contribute to environmental pollution.

.....

.....

.....

.....

2) Why is it crucial to address water pollution?

.....

.....

.....

.....

.....

Activity 1

List various activities that you perform everyday and analyse if they impact the environment positively or negatively.

7.4 MASS MEDIA AND ENVIRONMENT

Across a wide array of disciplines, people have been interested in understanding and harnessing mass media's influential power and scope for a long time. Because of this power of media to reach a large audience in a small amount of time, mass media is a tool that is incredibly good at rallying the support of a variety of diverse people at once. The media can also attract such a large audience because it simultaneously approaches people as the diverse beings that they are (Bennett & Entman, 2001).

The rapid expansion and breakthroughs in the arena of science and technology have ushered humankind into a new age. Unfortunately, the same advancement has had a devastating effect on nature itself. At this juncture, media plays a crucial role in forming public opinion and influencing policy decisions. An ordinary citizen gains confidence and capacity to become effective agents of change when aware of environment and development related issues.

7.4.1 Environmental Communication

Robert Cox defines environmental communication as “the pragmatic and constitutive vehicle for our understanding of the environment as well as our relationships to the natural world; it is the symbolic medium that we use in constructing environmental problems and negotiating society's different responses to them.”

Alexander Flor explains that “environmental communication has six essentials: knowledge of ecological laws; sensitivity to the cultural dimension; ability to network effectively; efficiency in using media for social agenda setting; appreciation and practice of environmental ethics; and conflict resolution, mediation and arbitration.”

Symbolic action of environmental communication serves two functions **pragmatic and constitutive**.

- 1) Environmental communication is pragmatic because it solves environmental problems by educating, alerting, collaborating, persuading and mobilising the masses. Communication plays a vital role in this area by helping citizens and organisations to achieve the goals towards solving environmental problems.

- 2) Environmental communication is constitutive because shaping our perceptions of the environment helps shape people's understanding of environmental issues.

7.4.2 Environmental Journalism

To be an environmental journalist, one must write about nature and understand scientific language and practice, knowledge of historical environmental events, the ability to keep abreast of environmental policy decisions, and the work of environmental organisations. S/he should have a general understanding of current environmental concerns, and the ability to communicate all of that information to the public in such a way that it can be easily understood, despite its complexity. With growing environmental pollution and industrialisation, environmental journalism came up to study and analyse all threats to the environment and ecology and convey these fears and their remedies to the people. The Bhopal Gas tragedy was the first instance that woke up the media and drove them to environmental journalism.

Dissemination of environmental facts: Technical jargon or generic terminology

Media is generally attracted to gloom and doom stories of climate change. But journalists must be exposed to the language and the concept of risks in covering climate science. With climate models becoming more powerful and sophisticated, media professionals are better equipped to quantify uncertainties and generate probabilistic climate projections, easily comprehensible to the layperson.

Media is often a target for lobby groups to amplify or underplay uncertainties around climate science for self-serving interests, leading to substandard reporting. Instead of this, journalists must enhance their competencies to handle such risk and be familiar with numbers and probabilities to formulate a more constructive narrative about climate change, i.e. reframing technical uncertainties cited in scientific terms into more comprehensive risk society measures. This would enable the people to perceive the problem, understand the risk and actively engage in public dialogue to seek sustainable solutions (Pidgeon and Fischhoff, 2011).

7.4.3 Functions of Mass Media towards Environment

The mass media plays a significant role in modern society. Mass media has certain essential functions to perform, which include influencing and moulding public opinion. In this modern knowledge society, media can play the following roles in spreading environment-related information:

Information Disseminator: Mass media play a significant role in shaping people's perceptions and awareness of environmental issues. People unknowingly imbibe various kinds of behaviours and attitudes from the media.

Agents of Change: The media and interpersonal interactions reinforce the message and bring about persuasion to change our attitudes, beliefs and behaviours. Though these are very difficult to change, mass media plays a significant role in helping to change political preferences and religious attitudes. Once an attitude is formed, the media functions to channel it in a specific direction.

Media as Stimulator: The mass media activates latent attitudes, prompting people to take action. It helps in mobilising the masses for collective action by

appealing to people's conscience to participate in various environmental conservation and support activities, e.g. using paper bags or jute bags instead of Polybags, etc.

Facilitator of development

Media coverage is crucial for any environmental issue to enter into the arena of public discourse and become a part of the political process. Civil Society Interest groups and non-governmental organisations (NGOs) working on environment-related issues can cultivate public opinion on issues of concern with the help of media.

Messenger to policymakers: Along with centre-staging the environmental issues into the hub of debates and discussions, media also tries to triangulate the opinions of different stakeholders. It works as a bridge by amplifying the popular opinions of people and communities across geographies and shares them with policymakers and vice-versa. This plays a significant role in encouraging governments to devise environment-friendly policies.

Across the world, rigorously covering and following upon events like the Bhopal Gas tragedy in 1984; and snowballing the discussion on the facts shared by the Academy Award-winning documentary on climate change, *An Inconvenient Truth*, in 2006, media has tremendously helped in creating a public opinion about significant environmental issues.

7.4.4 Media Platforms and Environmental Awareness

Media play an important role in influencing the attitudes of the public towards the environment. The media's role in increasing environmental awareness of the population is an enormous one as it reaches a vast percentage of India's complex society. Following are some common media platforms:

Newspapers and Magazines: Newspapers have always been a profound source of information about daily events with the morning cup of tea. They act as a source of motivation, for example, when they educate farmers about introducing organic farming, new agricultural technologies or sensitising them about the negative consequences of pesticides, stubble burning, etc. As a result of such exposure and subsequent public pressure, local authorities, governments, industries and other stakeholders are often forced to rectify their practices, to strongly enforce laws and regulations, and to abandon development projects if their environmental and social costs outweigh benefits (both organisational and societal).

Similarly, the environmental magazine "Down to Earth" covers various environment-related topics and their scientific background.

Radio: Radio is an affordable, most common medium of information and its signals cover almost the entire nation. It is noteworthy that the Ministry of Environment & Forests used to broadcast two weekly programmes on the environment, "Kinare-Kinare" and "Aao Dilli Savaren" on Delhi FM. At the national level, the news on environmental aspects is scant, and if they are broadcast, they are most often at the regional level.

Community Media: With the integration of media with community-based initiatives, community radio and video have opened doors for the remote

population to participate and initiate dialogues regarding their regional environmental concerns. Community media also gives them the freedom to deliberate on environmental protection activities to conserve and regenerate natural ecosystems. Henvallvani Community Radio (Chamba, Uttarakhand) has extensively produced programmes on the environment and climate change in the region.

Television: Studies indicate that television, by its audio-visual elements, enables a greater retention tendency. As a result, the government is increasingly interested in allocating prime time slots to environmental programmes on television. Presently, documentaries that revolve around the environment attract fewer viewers, maybe because of the academic or obscure manner in which they are presented. Mainstream channels such as the Discovery Channel, National Geographic and Animal Planet broadcast exclusively on endangered species and wildlife, sea life, other ecological aspects, etc. Programmes like “Virasat”, “Race to Save the Planet”, quiz show “Terraquiz”, “Earth” was telecast by the Ministry of Environment & Forest in collaboration with Doordarshan. In addition, BBC’s “Earth Report” offers exciting pieces of information on the environment. In fact, reruns of “The New Adventures of Captain Planet” on Cartoon Network catered to children’s younger population. They sensitised them about deforestation, pollution, poaching and other environmental hazards.

New Media: The Internet’s exponential reach and easy accessibility make it the prime resource for the global population to seek information about climate change, environmentalism, and how to be green and eco-friendly. Nowadays, Internet services are more frequently utilised for environmental awareness among people to engage in public dialogue almost instantaneously. Social media sites such as Twitter, Facebook, etc., share news, information, and articles and are thus, most resourceful for concerned “netizens” to keep abreast of the environmental issues. The Internet has also created spaces for convergence of traditional media with new media to produce an eclectic and multifaceted resource for people to gain indigenous knowledge about environmentalism. Furthermore, with the advent of mobile applications using different strategies to provide people with a sense of ownership of the planet, the search engine “Ecosia” ensures its users spend 100% of its profits on planting trees on suggested locations.

The following section provides information about prominent Indian Environmental Movements, discusses the initiatives taken by Indian media, and elaborates the functions of media advocacy for environmental projects.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Define environmental communication.

.....

.....

.....

.....

.....

- 2) Enlist the five functions of mass media towards environment.

.....

.....

.....

.....

.....

7.5 ENVIRONMENTAL MOVEMENTS

The environmental movement is a diverse scientific, social, and political movement for addressing environmental issues. Environmental movement is a type of “social movement that involves an array of individuals, groups and coalitions that perceive a common interest in environmental protection and act to bring about changes in environmental policies and practices” (Tong, Yank 2005).

According to Rootes, Christopher (1999), “The environmental movements are conceived as broad networks of people and organisations engaged in collective action in the pursuit of environmental benefits. Environmental movements are understood to be very diverse and complex, their organisational forms ranging from the highly organised and formally institutionalised to the radically informal, the spatial scope of their activities ranging from the local to the almost global, the nature of their concerns - from single issue to the full range of global environmental concerns. Such an inclusive conception is consistent with the usage of the term amongst environmental activists themselves and enables us to consider the linkages between the several levels and forms of what activists call ‘the environmental movement’”

Environmentalists advocate the sustainable management of resources and stewardship of the environment through changes in public policy and individual behaviour. The environmental movement is represented by a range of organisations - from International to grassroots.

7.5.1 Environment Conservation Movements in India

Table 7.1: Environmental Conservation movement in India
(compiled by the author)

Name	Year	About the movement
Chipko Movement	1973	Chipko movement was spearheaded by Chandni Prasad Bhatt and Sunderlal Bahuguna in the villages of Garhwal Himalayas, Uttarakhand. The movement was carried on by local community members, especially women, against mindless deforestation. They registered their protest by hugging trees when the woodmen came to axe them.

The Silent Valley Project	1978	Save Silent Valley was a social movement aimed at the protection of Silent valley, an evergreen tropical forest in the Palakkad district of Kerala, India. It was started to save the Silent Valley Reserve Forest from being flooded by a hydroelectric project.
Jungle Bachao Andolan	1980s	The Jungle Bachao Andolan took shape in the early 1980s when the government proposed to replace the natural Sal forest of Singhbhum District, Bihar (now Jharkhand) with commercial Teak plantations. Historically, this region has seen rebellion, victory and loss in the tribal communities struggle to live and work in their own forests.
Navdanya Movement	1982	Navdanya promotes biodiversity conservation, biodiversity, organic farming, the rights of farmers, and the process of seed saving. Navdanya has led the national and international movement for bio-safety and against the dangers of Genetically Modified Organisms (GMOs) in agriculture.
Bhopal Gas Tragedy	1984	The Bhopal gas tragedy was a gas leak incident in India, considered one of the world's worst industrial disasters. It occurred on the night of 2–3 December 1984 at the Union Carbide India Limited (UCIL) pesticide plant in Bhopal, Madhya Pradesh. Scores of environmentalists, civil society and community based organisations are working together since then to bring justice to the ones who died in this tragedy by lobbying for tougher environmental laws and regulations.
Narmada Bachao Andolan	1985	Narmada Bachao Andolan is a social movement against a number of large dams being built across the Narmada river which flows through Madhya Pradesh, Gujarat and Maharashtra. Tribals, farmers, environmentalists and human rights activists carried out their protest by doing hunger strikes and garnering support from noted film and art personalities, together with its leading spokespersons Medha Patkar and Baba Amte.

7.5.2 “Save our Tigers” Campaign

Decline in the number of tigers in India was highlighted by a media campaign. The current population of tigers in India constitutes over 50% of the world's tiger population. The coming years will be extremely crucial for the long term survival of the tiger. In 2004, environmentalists were shocked when not a single tiger could be spotted at the popular Sariska wildlife reserve in Rajasthan. It was seen as the biggest crisis in India's conservation history and led to initiation of campaigns to save the tiger. India has struggled to stop the rapid decline of tiger population from an estimated 100,000 at the beginning of the 20th century, in the face of poachers, international smuggling networks and loss of habitat.

NDTV and telecom operator Aircel launched a media campaign called “Save Our Tigers”. The campaign was a social campaign to create awareness about the alarmingly dwindling population of tigers across the country, promote tiger conservation efforts and save tigers from extinction. The campaign is spearheaded by Sanctuary Asia, India's premier wildlife magazine and the Wildlife Conservation Trust with film star Amitabh Bachchan being the face of the campaign.

The campaign provided a platform to tiger conservationists to raise this issue, engage key stakeholders in discussions and provide concerned citizens an opportunity to voice their opinions and contribute to the cause. In a bid to save tigers, this campaign ignited people across the country to come together for collective action. There were marches, cycle rallies and signature campaigns demanding from the government to act before it's too late.

7.5.3 Media Advocacy in Environmental Organisations

Strategic uses of news created through mass media can promote public debate and generate community support for changes in community norms and policies on environmental issues in India and other nations. It is necessary for such platforms to devote an adequate amount of airtime and space to produce an effect on people, who then take the desired action. Many studies establish that members of the general public are the ultimate decision-makers and opinion leaders in our society. Dialogues that take place within families and social networks we experience in our everyday lives, shape our attitudes, beliefs, norms, and practices. Therefore, leveraging the potency of media advocacy in attaining sustainable development goals is key to healthy and prosperous communities. Building on this backdrop, many organisations have evolved their advocacy units to address the most complex environmental issues. We briefly discuss the endeavours of a few environmental organisations:

- A) The Palm Oil Controversy:** Greenpeace's strategic use of social media to urge Nestle to stop buying palm oil for its infamous chocolate bar 'KitKat' was a breakthrough. The palm oil used in manufacturing this chocolate (and many other household products) is an endangering threat to Orangutans, and Nestle used to avail the ingredient from Sinar Mas. This company deliberately smoked out Indonesian rainforests to make space for palm oil plantations. Greenpeace demanded Nestle to refrain from using palm oil and used Facebook, Twitter, YouTube, and even the blogosphere to its aid. The environmental organisation also wrote an open online letter to the Public Relations team of Nestle when the latter began to avail palm oil from an intermediary (Cargill) to avoid direct repercussions.

- B) Basin Report Card Initiative:** Basin report cards provide a platform for gathering information about basin health, synthesised and delivered via a public platform, and people are empowered and can take action. Developed from the ground-up and rooted in science, report cards identify what is most important to the diverse water users in a given basin, creating a shared understanding of the basin's health to foster a shared vision for its future. World Wildlife Fund (WWF) seeks to empower stakeholders worldwide to develop and effectively use credible, locally owned report cards in their basins, fostering sustainable water management across basins around the world. It provides visual elements such as photos, maps and conceptual diagrams to the media to disseminate among government entities, the private sector and the public-at-large to initiate dialogue and inspire positive change. Further, WWF has designed a simple online quiz for users to play and determine the health of any freshwater basin. Results of this quiz suggest possible actions to reduce water pollution.
- C) The Energy and Resource Institute:** In addition to a vast body of scientific knowledge on energy and environmental issues, TERI produces films portraying stories of change, which capture the human side of the story through powerful images and voices from across India that brings out the many struggles and solutions related to sustainable development.
- D) Center for Science and Environment:** CSE organises media fellowships for media professionals to develop systematic content on various thematic areas such as good food, intelligent cities, minerals use, pollution and renewable energy. They conduct capacity building workshops for journalists representing different platforms of television, radio and newspapers. By this training, the fellows are groomed to write data-driven, thought-provoking stories to influence the general public and enhance their competence to determine the course of action beneficial to society at large.
- Other organisations include Central Pollution Control Board (India), United Nations Environment (UNE), Wildlife Protection Society, Indian Environmental Society, Center for Environmental Studies, Development Alternatives (DA), Navdanya Trust, Water Aid, Water for People, among many others.
- E)** In addition, campaigning organisations such as Jhatkaa.org are standing up against large-scale corporations like Hindustan Unilever and issuing online petitions using innovative methods. To create awareness about mercury poisoning in Kodaikanal, Jhatkaa.org roped in a Chennai-born rapper to write a parody song. The song's video took an undisguised jab at Unilever for its failure to clean up mercury contamination for over 14 years or compensate workers affected by its thermometer factory, despite promising numerous claims about its social responsibility towards the environment. With more than 3.1 million hits on YouTube, it stirred a mass movement by reinforcing the message - 'Kodaikanal won't (suffer)'; this ensued desired action by the corporation.

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

- 1) Briefly discuss two environmental movements with an objective of preservation of Indian forests.

.....

.....

.....

.....

.....

- 2) Give two examples of media advocacy.

.....

.....

.....

.....

.....

7.6 INTERNATIONAL ENVIRONMENTAL AGREEMENTS

Countries around the world have formulated Environmental Agreements at the global, national and local levels. Each country has laid down specific laws and legislations, which are unique to their region. International Agreements have been formulated to collectively address the issues of Environment. Some of the important Environmental Conventions and Protocols are tabled below, while many more are being formulated.

7.6.1 Key International Conventions and Protocols

The following table lists the Key international Conventions as protocols.

S.No.	Conventional Protocol	Objectives	Year
1)	Montreal Protocol	The Montreal Protocol on Substances That Deplete the Ozone Layer specified that the production and consumption of compounds –chlorofluorocarbons (CFCs), halons, carbon tetrachloride, and methyl chloroform, that deplete the ozone layer in stratosphere, be phased out.	1989
2)	Base Convention	The Base Convention on Tran boundary Movement of Hazardous Wastes and their Disposal was adopted in response to concerns about the toxic waste from industrialised countries being dumped with economics in transition. It aimed to:- • minimise the generation of hazardous wastes in terms of quantity and hazardousness;	1992

		- dispose them off as close to the source of generation as possible; - reduce the movement of hazardous wastes.	
3)	Convention on Biological Diversity(CBD)	The three objectives of the CBD are: conservation of biological diversity; sustainable use of its components and; fair and equitable sharing of the genetic resources.	1993
4)	UNFCCC	The United Nations Framework Convention on Climate Change (UNFCCC) is the foundation of global efforts to combat global warming. The objective of this Convention is stabilisation of greenhouse gas concentration in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system.	1994
5)	Kyoto Protocol	In December 1997, more than 160 nations met in Kyoto, Japan, to negotiate binding limitations on greenhouse gases for the developed nations, as per the objectives of the Framework Convention on Climate Change of 1992. The outcome of the meeting was the Kyoto Protocol, in which the developed nations agreed to limit their greenhouse gas emission, relative to the levels emitted on 1990	1997
6)	Stockholm Convention on POPs	The Stockholm Convention on Persistent Organic Pollutants (POPs) is a global treaty adopted to protect human health and the environment from POPs. The Convention seeks the elimination or restriction of production and use of all intentionally produced POPs (i.e. Industrial chemicals and pesticides).	2001
7)	Nagoya Protocol	The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilisation to the Convention on Biological Diversity is an international agreement which aims at sharing the benefits arising from the utilisation of genetic resources in a fair and equitable way– a) by appropriate access to genetic resources; b) by appropriate transfer of relevant technologies, taking into account all rights over those resources and to technologies, and; c) by appropriate funding, thereby contributing to the conservation of biological diversity and the sustainable use of its components.	2010

7.6.2 U.S. India Partnership on Climate Change

As the world's third largest carbon emitter, India is a top priority for the U.S. Presidential Global Climate Change Initiative. The U.S. Government is supporting clean energy in India through the inter-agency Partnership to Advance Clean Energy (PACE) effort, which includes USAID; the U.S. Departments of Commerce, State, and Energy; the Export-Import Bank; the Overseas Private Investment Corporation; and the Trade and Development Agency. The goal is to accelerate India's transition to a high-performing, low-emissions and energy-secure economy as part of the U.S.-India partnership on climate change.

Check Your Progress Exercise 4

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

- 1) What are the steps taken at the international level to collectively address the issues of Environment?

.....

.....

.....

.....

.....

- 2) Throw some light on U.S. India partnership on Climate change.

.....

.....

.....

.....

.....

7.7 LET US SUM UP

In this unit, you learnt about various risk factors for the environment, such as air pollution, chemical wastes, climate change, water scarcity, etc. The differences between environmental communication and environmental journalism and the role of various mass media in creating awareness about environmental issues were explained. You were also exposed to several movements initiated in India over the years to help protect the environment. Towards the end, various international environment agreements that have been formulated to address the issues of Environment and create a better living for us were outlined

7.8 REFERENCES AND FURTHER READINGS

Acharya, K. & Noronha, F. (Eds). (2010). *The Green Pen: Environmental Journalism in India and the South Asian Region*. New Delhi, India. SAGE

Bennett, W. L. & Entman, R. M. (Eds). (2001). *Mediated Politics: Communication in the Future of Democracy*. New York, NY: Cambridge University Press.

Cox, R. (2010). *Environmental Communication and the Public Sphere* (2nd ed.). Thousand Oaks: Sage Publications

Flor, A. (2003). *Environmental Communication*. Diliman, Quezon City: University of the Philippines-Open University

Further Readings

Agarwal, Bina (2007). 'The gender and environment debate: Lessons from India.' In Mahesh Rangarajan (Eds.). *Environmental Issues in India: A Reader*. New Delhi: Pearson India.

Braidotti, R., E. Charkiewicz, S. Hausler, S. Wieringa (1994). (Eds.). *Women, the Environment and Sustainable Development: Towards a Theoretical Analysis*. London: Zed Books with INSTRAW.

Ortner, Sherry, B. (1972). 'Is female to male as nature is to culture?'. *Feminist Studies*, 1(2), 5-31.

Sarin, Madhu (2001). 'Empowerment and disempowerment of forest women in Uttarakhand, India'. *Gender Technology and Development*. 5(3).

Mahesh Rangarajan (Eds.). *Environmental Issues in India: A Reader*. New Delhi: Pearson Longman.

Salleh, Ariel (1993). 'Class, race and gender discourse in the ecofeminism/ deep ecology debate'. *Environmental Ethics*, 15, 225-44.

Shiva, Vandana (1988). *Staying Alive: Women, Ecology and Survival in India*. New Delhi: Kali for Women.

Warren, Karen & N. Erkal (1997). *Ecofeminism: Women, Culture and Nature*.

Warren, Karen (2000). *Ecofeminist Philosophy: A Western Perspective on What it is and Why it Matters*. Rowman and Littlefield.

Online Resources

<https://www.youtube.com/watch?v=RAQxalkWPSw>

<https://www.youtube.com/watch?v=a8JaaeRU9DA>

https://www.youtube.com/watch?v=J_0sUSt250Y

7.9 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Use of fossil fuels which degrade quickly; rising fuel emissions cause air pollution, and pose a threat to environment and human health.
- 2) The demand for water – for drinking, sanitation, farming and energy production, among many other uses – grows with the rise in population. At the same time, human activity and climate change disrupt the natural water cycles, acting as a stressor for freshwater ecosystems. Pollution, infrastructure development and resource extraction pose additional challenges.

Check Your Progress Exercise 2

- 1) It is the pragmatic and constitutive vehicle for our understanding of the environment as well as our relationships to the natural world; it is the symbolic medium that we use in constructing environmental problems and negotiating society's different responses to them
- 2) Five functions of mass media towards environment are: information disseminator, agent of change, media as stimulator, facilitator of development, and messenger to policy makers.

Check Your Progress Exercise 3

- 1) Chipko Movement - in the villages of Garhwal Himalayas, Uttarakhand. People protested against deforestation by hugging trees when the woodmen came to axe them. The Silent Valley Project - in the Palakkad district of Kerala, India. It was started to save the Silent Valley Reserve Forest from being flooded by a hydroelectric project.

- 2) Jhatkaa.org ; against mercuric poisoning in Kodaikanal
Greenpeace; campaign against Nestle's use of Palm oil

Check Your Progress Exercise 4

- 1) Countries around the world have formulated Environmental Agreements at the global, national and local levels. Each country has laid down specific laws and legislations, which are unique to their region. International Agreements have been formulated to collectively address the issues of Environment.
- 2) The U.S. Government is supporting clean energy in India through the inter-agency Partnership to Advance Clean Energy (PACE) effort.



UNIT 8 ECONOMY AND FINANCE

Structure

- 8.0 Introduction
- 8.1 Learning Outcomes
- 8.2 Economic and Finance Reporting
 - 8.2.1 Economic Reporting
 - 8.2.2 Financial Reporting
- 8.3 Avenues of Investment
 - 8.3.1 Share Market
 - 8.3.2 Rising interest in Mutual funds
 - 8.3.3 Expanding Middle class
 - 8.3.4 Increase in business and financial Journalism
- 8.4 News and their Sources
 - 8.4.1 Reporting Share Prices
 - 8.4.2 Caution
 - 8.4.3 Sensex and Nifty
- 8.5 Malpractices in Reporting
- 8.6 Let Us Sum Up
- 8.7 Keywords
- 8.8 References and Further Readings
- 8.9 Answer to Check Your Progress Exercises

8.0 INTRODUCTION

India is now the world's sixth largest economy at 2.6 trillion dollars. India is targeting to be a five trillion dollar economy by 2022. According to the 2019 global Multidimensional Poverty Index (MPI) released by the United Nations Development Programme (UNDP), more than 271 million people in India moved out of poverty in the decade since 2005-15. Per capita income has also increased significantly and this economic growth has created various changes in the quality of an average person's life in the country. At the time of independence life expectancy in India was 32 years; it has now increased to 68.7 year.

The importance of money in our society is steadily increasing. A large section of the growing middle class in India, with surplus income, has started taking keen interest in making money through shares and mutual funds market. This has led to the emergence of several new financial newspapers and television channels in English and other major Indian languages and greater interest and consumption of economic and financial news.

In this Unit, we shall discuss different aspects of economic and financial reporting. We shall look at the some avenues of investment such as share market, mutual funds, and how they have impacted financial markets. We shall also discuss some malpractices in financial reporting and highlight the precautions that you need to take while reporting on share markets.

8.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- discuss the need and importance of economic and financial reporting;
- differentiate between economic and financial reporting;
- examine the malpractices in financial reporting; and
- explain the terms used in economic and financial reporting.

8.2 ECONOMICS AND FINANCE REPORTING

Economics and finance are interrelated, but some unique features differentiate one from the other. Finance is a subset of economics that studies the financial markets. The branches of economics are Macro Economics and Micro Economics.

Macro Economics deals with performance, structure, behaviour, and decision-making in an economy as a whole. This includes regional, national, and global economies. It is a study of aggregated indicators such as Gross Domestic Product (GDP), unemployment rates, national income, price indices, and the interrelations among different economic sectors to understand better how the whole economy functions. It also develops models that explain the relationship between national income, output, consumption, unemployment, inflation, savings, investment, international trade, and international finance.

Micro Economics concerns itself with the behaviour of individual markets, such as the markets for oranges, cable television, electronics, etc. Micro Economics is essential for local governance, business and personal financing, specific stock investment, etc.

8.2.1 Economic Reporting

Economic reporting means news about the production, consumption and distribution of services and goods. It influences policy-making at the national level and provides guidance for the same. Economics deals with trade across the entire spectrum, unlike markets. It deals in policy in regard to taxation, government spending, etc.

Many eagerly wish to know the details of shopping and the emergence of big shopping malls affecting small businesses. In Britain, providing licenses to new big departmental stores to open 24 hours led to the closure of many corner shops. In the past few years, labour unions have lost their influence. People would like to know how it impacts our economy and the life of industrial labourers in our country.

Economics is a vast subject. It includes many areas: liberalisation, privatisation, globalisation, poverty, human capital, rural development, employment, environment, and sustainable development. Reserve Bank of India's periodic report is an index of the state of the economy.

An area of potential economic news not adequately covered by media is the rural, farm and agriculture sector. In our country, there is no income tax on agricultural income. It is often argued that wealthy farmers have misused this

benefit with significant income to evade taxes. Moreover, farmers also cry for loan waivers, and political leaders succumb to such pressures because of vote bank politics. It is expected of a reporter to explain how it adversely affects our economy and public sector banking. This system is making farmers more prosperous in some areas. However, in some other parts of the country, farmers commit suicide because of their loan burden. Newspapers are expected to explain this situation to their readers.

8.2.2 Financial Reporting

Financial reporting primarily deals with share markets, mutual funds, bullion prices, trading in consumer goods, company news, new products, etc. Financial journalism in India has undergone a revolutionary change in recent years because of the changes in Indian financial markets.

Before the advent of online share trading in India and 24x7 business channels, our share market was not so quickly impacted by the happenings in stock exchanges of New York, London, Tokyo, Hong Kong and Singapore. Now a financial reporter has to be alert about developments in other major stock exchanges and foreign currency markets. Even crude oil prices in the international market impact our share market.

Sometimes our Sensex keeps on rising but small, and mid-cap share indexes show a negative trend. A reporter has to explain the reasons for this scenario. Readers expect a financial reporter to tell them how any change in interest rates declared by the RBI would affect the share and debt market of the country.

Activity 1

Take any two business newspapers (say, Economic Times, Business Standard), compare their pages of share market rates. List how they present share rates differently.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is the difference between economics and finance?

.....

.....

.....

.....

.....

2) What are the subjects covered in economics?

.....

.....

.....

.....

3) What are the major areas covered in financial news?

Economy and Finance

.....

.....

.....

.....

.....

8.3 AVENUES OF INVESTMENT

If a person wishes to invest his/her disposable income there are usually the following financial avenues where the investment can be made:

- Stock market
- Bank, and post office fixed deposits
- Bullion which includes gold and silver
- Real estate.

Now days, people are also investing in mutual funds. Mutual fund companies invest in share market and debt market. Debt instruments are of private sector as well as public sector. At present, mutual funds are a major source of investment in stock market. However, except for fixed deposits, profits on other investments are not assured, as there is a risk of financial loss, depending on the market conditions.

8.3.1 Share Market

Mass participation of retail investors in share market started in the 1980's with the entry of Dhirubhai Ambani. In fact, Dhirubhai Ambani can be called the father of modern share markets in India who made common people interested in his company's shares and gave them good dividends. Other important changes in stock markets took place with the introduction of online trading in 1995 and the establishment of online depository of shares in 1996. With this, one can now buy and sell shares through the Internet.

8.3.2 Rising Interest in Mutual Funds

The Indian mutual fund industry started with the enactment of an Act in 1963. The first scheme launched by Unit Trust of India (UTI) was Unit Scheme 1964, popularly known as US 64. It was very popular among common people. But in the year 2001, US 64 posted a great loss and investors lost heavily. Mutual fund industry then made a big leap forward after 2003 when the Unit Trust of India Act, 1963, was repealed.

The mutual funds in India again got a big boost due to globalization. Now a day, rise and fall of mutual fund business in India is directly linked with the ups and downs in the share markets. Mutual Fund companies offer equity funds, debt funds, balanced funds, etc. There are gold funds too which are in unit-form and not in the physical form. Multi Commodity Exchange, established in 2003 in India, is also important for financial reporting. The Multi Commodity Exchange

offers online trading of several agricultural commodities including oils and soya-bean, and some ferrous and non-ferrous metals. All these developments have greatly influenced the financial markets and financial journalism.

8.3.3 Expanding Middle class

During the past few years the number of share-holders in public limited companies in India has increased many folds. However, this does not mean that the control over companies owned by Tatas, Birlas, Ambanis, and other industrialists, has gone into the hands of the middle classes. Similarly, a large number of public sector undertakings, including the State Bank of India, Indian Oil Corporation, ONGC and Coal India, have floated shares for the general public. Due to all these factors, the stakes of readers of newspaper and viewers of television have increased many times in the financial markets of the country.

8.3.4 Increase in Business and Financial Journalism

All leading newspapers of India have increased the number of their business pages in recent years. Even in general television news bulletins, reporting of rise and fall of Sensex has become a routine story like a weather report. The phenomenal increase in general readership of the Economic Times, Financial Express, Business Standard, Business Line, Mint and other business magazines has inspired many newspapers to give greater coverage to business and financial news.

Some of these newspapers and magazines have launched their editions in regional languages as well. Apart from this, there are magazines in Hindi, English, Gujarati, etc. which are fully devoted to financial markets. Gujarati language newspapers, Sandesh and Gujarat Samachar publish a weekly column on bullion markets. Moreover, more than half a dozen business television channels have also come up. These include NDTV Profit, ET NOW, CNBC TV 18, UTV, Zee Business, Bloomberg, Awaaz, etc.

Today, specialising in business and financial journalism has become an attractive area for young journalists. A financial reporter should have an excellent interpretative mind. S/he should be able to explain why stock market is falling, why people are taking more interests in fixed deposits than in stock market, and why mutual fund market is rising or falling, and so on so forth.

Business journalism also analyses and interprets the economic changes that take place in a society. It may include anything from personal finance to share market and company news; to commodity market and bullion prices, and to business at the local market and shopping malls.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

- 1) Mention two important changes that took place in the stock market during the mid 90s.

.....

.....

2) What skills a financial reporter should have?

.....

.....

.....

.....

.....

3) List a few public sector companies whose shares could be purchased by the public.

.....

.....

.....

.....

.....

8.4 NEWS AND THEIR SOURCES

A business reporter must be fully conversant with the public and private agencies from where information can be obtained. Much of the routine news reaches media houses on its own periodic reports of RBI, government and industry - sent directly by their respective public relations departments. Press conferences are also organized by the business houses from time to time. Information is shared about the annual reports of companies, their quarterly and half yearly results, how much advance income tax they have deposited, their new products launched, etc. Share and commodity market rates and bullion rates also come in the category of routine news. Mutual fund unit rates are announced every day by different fund houses. Similarly, foreign exchange rates are a routine matter. RBI's periodic reports indicate the economic health of the nation. Mumbai being the financial capital of India, maximum financial news originates from there. The financial editor selects only that news which is of interest to his readers or viewers which may vary from place to place. For example, all leading Gujarati newspapers publish share market rates, whereas for Hindi dailies of Jharkhand, these rates may not be that important.

8.4.1 Reporting Share Prices

It is a challenge for a newspaper organisation as well as TV channel to provide space or time to various share transactions in the country. The same problem is about the growing number of mutual fund rates. At times the business page editor has to reduce the font size of letters to accommodate maximum share transactions and mutual funds rates on the page. Every editor must exercise judgment as to what items to be included in the limited space at his disposal. Some newspapers

carry mutual fund rates on weekly basis, that too of only important mutual fund units. Similarly, TV channels give rates of important shares only.

8.4.2 Caution

Any wrong business story may cause a great deal of financial damage to its readers and viewers. Therefore, a business or financial reporter has to be very accurate. S/he should not use any financial or economic term in a story whose meaning is not clear. A reporter should have good knowledge of terms like Sensex, Nifty, blue chip companies, dividend, debentures, bonds, debt funds, balanced funds, bonus shares, preference shares, FMPs, etc. In subsection 4.8, we have given an extensive glossary of the terms which every budding financial and economic journalist should be familiar with.

A common problem with financial reporters, particularly TV reporters, is that stories come with such speed that they have little time to comprehend their implications properly. News media works under-pressure and at times, reporters are unwittingly used by smart PR strategists in an increasingly competitive media market.

8.4.3 Sensex and Nifty

Sensex, also called BSE 30, is the market index consisting of 30 well- established and financially sound companies listed on Bombay Stock Exchange (BSE). Thirty companies are selected on the basis of the free float market capitalisation.

Nifty, also called NIFTY 50, is the market index consisting of 50 well- established and financially sound companies listed on National Stock Exchange of India (NSE). Located in Mumbai, NSE was established in 1992 as the first demutualised electronic exchange in the country.

Activity 2

Go through a financial newspaper or magazine of your choice. Make a list of 30 shares of Sensex and 50 stocks of Nifty. Identify any stock in Sensex list which is not in Nifty.

Check Your Progress Exercise 3

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Name the two main share markets of the country.

.....

.....

.....

.....

2) What is the difference between Sensex and Nifty?

.....

.....

.....

.....

.....

.....

.....

.....

8.5 MALPRACTICES IN REPORTING

Wherever finances are involved, some people try to find out ways to make money through questionable practices. Financial reporting is also not free from such malpractices. In his article ‘Crisis in Indian Media’ published in the PCI Review, Shyam Khosla, a senior journalist, observed:

“Some publishers and TV channel owners unashamedly claim that they are there to serve the market. The financial media is notoriously corrupt. Costly gifts and company shares are given to financial reporters in lieu of favourable stories for the company concerned. This is no longer a secret. Unfortunately, no one speaks out against the corporate world that has largely remained unexamined. Is it because they are big advertisers?”

Similarly, in an article ‘Share market manipulation by big media houses curbed’ by Himanshu Chatterjee and Anil Mahatme (2010) stated that some media houses indulge in promoting shares of particular companies. Securities Exchange Board of India (SEBI) has made it mandatory for media groups to disclose in their news reports their stakes in private companies, which are

being reported on. The Press Council of India has also expressed its concern over practice of many media groups making private treaties with companies which are listed or which are going to offer a public issue of their shares. Through this arrangement, media groups - both print as well as electronic, help in promoting shares of particular company through advertisements, news reports, advertorials, etc. In 1996 the Press Council of India counseled reporters, financial journalists, and media establishments to refrain from receiving any gifts, grants, concessions, facilities, etc. either in cash or kind which are likely to compromise free and unbiased reporting on financial matters.

Financial journalism provides enough scope for investigative reporting. Sucheta Dalal grabbed that opportunity in 1992 to expose Harshad Mehta’s scams in Mumbai share market. Later Harshad Mehta was imprisoned and died in a Mumbai jail in 2001. Dalal co-authored a book *The Scam: Who Won, who Lost, who Got Away*, which investigates the role of several figures in the April 1992 crash of the share market in India. In 2006, Government of India awarded Padma Shri to Sucheta Dalal in recognition of her journalistic contribution.

The share market business has enough scope to grow in India. At present, this business is largely concentrated in cities like Mumbai, Pune, Nasik, Ahmadabad, Rajkot, Surat, Vadodara, Bangalore, Kolkata, Hyderabad, Kochi, Chennai, Chandigarh, Delhi and its capital region including Noida and Gurugram. In large parts of India, there is less activity as far as share business is concerned.

Seeking explanations

We are living in an era of investigative, analytical and interpretative journalism. When Sensex falls a reporter should give reasons why it has fallen. When rupee falls against dollar, the reporter needs to explain reasons for the fall. Without reason or explanation, a news story is incomplete as readers want to know reasons for each rise and fall in Sensex. Curtis D. MacDougall in his book Interpretative Reporting (Sixth Edition) said:

“Whatever the truth may be, the financial reporter has the responsibility to seek reasons for important fluctuations. ... The alert financial reporter constantly asks “why?”, especially when any form of behaviour seems to become widespread, (such) as investing capital abroad, establishing branch banks, splitting stock, purchasing government bonds ... and so forth.”

The financial reporter should have elementary knowledge to read stock quotations.

MacDougall further said: “Because of the injury that might otherwise result, the business-financial page reporter must be deadly accurate. Among other precautions he must observe is not to use business or financial terms unless he thoroughly understands their meaning. He should not hesitate to ask news sources to define and explain unfamiliar words and expressions.”

Check Your Progress Exercise 4

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What are Blue Chip Stocks?

.....

.....

.....

.....

.....

2) What is devaluation?

.....

.....

.....

.....

3) What is Net Asset Value (NAV)?

.....

.....

.....

.....

.....

8.6 LET US SUM UP

In this Unit we discussed various aspects of economic and financial reporting. It was explained that economics and finance are interrelated, but are different too as economics is broader in nature; and finance is a subset of economics. In India, the 40 million strong middle class with surplus income is growing fast and as a result, people are becoming more aware and interested in investing their surplus income. Therefore, their interest in economic and financial news has increased. This has led to the growth of a specialized media, both in print and audiovisual, exclusively devoted to economic and financial news.

We discussed various avenues of investment such as share market and mutual funds among others. Share market, mutual funds, company news, new products are covered by financial and business reporters. We also discussed the precautions that are necessary while reporting such news.

A financial reporter must know what Sensex and Nifty are and how their shares are different from other stocks. A reporter should always explain why Sensex is falling or rising and why rupee is falling against US dollar. If some development is taking place, the reporter should give reason for that as the reader wants to know it. Since any wrong business story may cause a lot of financial damage to media users, a journalist must be very accurate and factual in business stories.

The scope of economic and financial reporting is growing and it has a great future because it is directly related to the development of the country.

8.7 KEYWORDS

The following are economic and financial terms which are helpful to a reporter in the field:

Abridged Prospectus: Abridged Prospectus means the memorandum as prescribed in Form 2A under sub-section (3) of section 56 of the Companies Act, 1956. It contains all salient features of a prospectus. It accompanies the application form of public issues.

Accidental Death Insurance: Accidental Death Insurance provides coverage in the event of death due to accidental injuries but not illness. In the event of death payment is made to the insured's beneficiary. And most of these covers provide for cases for bodily injury (e.g., the loss of a limb), where the insured receives a specified sum.

Accrual Basis: A method of accounting that allows revenues and expenses to be accrued, even if cash had not been received or paid during the accrual period.

Accrued Expenses: Expenses shown on the income statement but not yet paid.

Annual General Meeting (AGM): The shareholders meeting usually held at the end of each financial year, to discuss the previous performance and future outlook.

Annuity: An investment where the investor provides cash to the vendor, typically an insurance company, in exchange for the promise to pay a series of periodic, usually monthly, payments in return. An immediate annuity begins to pay back

right away; other annuities might not pay back for many years. The money invested in an annuity grows tax-deferred.

Annuitant: An individual receiving benefits under an annuity.

Asset Management Company (AMC): A Company registered with SEBI, which takes investment/divestment decisions for the mutual fund, and manages the assets of the mutual fund.

AMFI: Association of Mutual Funds in India or AMFI represents all Fund Houses as a group in terms of any discussion with SEBI, distributors or investor associations. It also aims to promote best practices among mutual funds. All mutual fund distributors need to be registered with AMFI.

Assets under management (AUM): Refers to the cumulative market value of investments managed by a mutual fund.

Asset: Any holding with monetary value such as stocks, bonds, real estate, gold and cash.

Arbitration: Procedure in which two parties agree to settle a claim dispute by accepting a decision made by a third party.

Audit: Examination and verification carried out by an outside agency (such as an accountancy firm or the tax authorities) of a taxpayer's books and accountants and/or the general accuracy of returns and declarations, either as a routine operation, or where evasion is suspected.

Auditor: An auditor is an individual who conducts an examination and verification of a company's financial and accounting records and supporting documents.

Budgetary deficit: A situation when the government's income and tax receipts fail to cover its expenditure

Balanced fund: A mutual fund scheme with an investment objective of both long-term growth and income, through investment in stocks and bonds. Generally 65% is invested in stocks and 35% in bonds, in order to provide access to relative stability of fixed income securities and the growth potential of equities.

Bear Market: A period of time during which securities prices are falling in the stock market.

Broker: A broker is basically a sales person who distributes stocks, bonds, or mutual funds. A broker is a licensed person authorized to receive commissions. Brokers are always affiliated with a brokerage company or broker-dealer network.

Bull Market: A distinctive time period, during which the prices of securities are rising, usually characterized by high trading volumes.

Bad debt: Debt which is unlikely to be paid. Bad debts may usually be treated as losses and written off against a reserve for such debts.

Balance of Payment: A summary of the international transactions of a country over a period of time including commodity and service transactions, and gold movements.

Balanced fund: Balanced fund include both equity and debt mutual fund schemes, with mostly 50-75 per cent in equity and the rest in debt.

Bank guarantee: Contract between a bank as guarantor and a beneficiary in which the bank commits itself to pay a certain sum under certain, specified conditions. Thus, a demand guarantee is one in which the bank agrees to pay against the simple written demand of the beneficiary.

Barter: The direct exchange of goods and/or services for other goods and/or services without the use of money and without the involvement of a third party. Barter is an important means of trade with countries using currency that is not readily convertible.

Benchmark: An unmanaged group of securities whose performance is used as a standard to measure investment performance. It is commonly known as a market index. Some well-known benchmarks are the BSE Sensex and NSE Nifty.

Blue Chip Stocks: Stocks of leading and nationally known companies that offer a record of continuous dividend payments and other strong investment qualities.

Bond: A debt security issued by a government entity or corporation, which generally pays a specified rate of interest, and plans to return the principal amount of the loan on the maturity date.

Broker or brokerage firm: A securities firm or a registered investment advisor affiliated with a firm. Brokers are the link between investors and the stock market. When acting as a broker for the purchase or sale of listed stock, the investment advisor does not own the securities but acts as an agent for the buyer and seller and charges a commission for these services.

Commercial paper: Short-term, unsecured, and usually discounted promissory note issued by commercial firms and financial companies often to finance current business. Commercial paper, which is rated by debt rating agencies, is sold through dealers or directly placed with an investor.

Commodity exchange: A commodity exchange is an association, or a company or any other body corporate organizing futures trading in commodities.

Compound interest: The ability of a fixed deposit in India to generate interest, which is then reinvested in order to generate further earnings, is known as compound interest.

Capital Gains/Losses: Net profit or losses from the sale of securities in the fund's portfolio. These are further classified as short term and long term capital gains. The holding period for each type of capital gains as defined in the Income Tax Act 1964 varies from instrument to instrument.

Commission: A fee charged by a broker or distributor for his or her service in the buying or selling of securities/ mutual fund units.

Compounding: Interest earned not only on the initially invested principal but also on accumulated interest during the period.

Consumer price index: The index compiled by a governmental agency which tracks the cost of living by following the change in prices of basic goods and services over time. This index measures inflation.

Credit rating: A measure indicating the bond issuer's credit worthiness or the company's ability to repay the loan. The bonds are rated by an independent rating agency such as CRISIL, ICRA, and CARE, etc.

Cash reserve ratio: A proportion of the total deposits and reserves of the commercial banks that is to be kept with the central bank (RBI: Reserve Bank of India) in liquid form. It is used to as a measure of control of RBI over the commercial banks.

Default: Failure to make repayment to the lender of the principal and interest on a debt on the schedule date.

Deficit financing: A situation where the expenditure of the government exceeds its revenue.

Devaluation: A fall in the exchange rate which reduces the value of a currency in terms of other currencies.

Demat Account: Electronic trading of shares required that shares be transferred electronically owing to which physical shares were dematerialised. One holds these shares in a demat account.

Dividend: An income distribution to shareholders that generally comes from the net profit or earnings of a corporation (or net income from a mutual fund).

ELSS: Equity Linked Savings Scheme or ELSS allows an investor to get the benefit under Sec 80C of the Income Tax Act.

Exit load: The charge levied at the time of exit is called exit load. This may be applicable in some schemes and for a certain period. One must check exit loads before investing in a mutual fund scheme.

Export duties: Taxes imposed on goods exported from a country

Face value: The value printed on the face of a stock, bond or other financial instrument or document. It is also the issue price of a mutual fund unit. Percentage dividends are calculated on face value.

FMP: Fixed Maturity Plan or FMP has a defined term after which the scheme ceases to exist. It is a closed ended fixed income scheme.

FMCG: Fast Moving Consumer Goods

Folio: A folio is an investor's unique mutual fund account. Much like a bank account number, a Folio number depict their holdings in the schemes of a fund house.

Fiscal policy: All the planned actions of a government in mobilizing financial resources for meeting its expenditures and regulating the economic activities in a country.

Gross Domestic Product (GDP): The total value of final goods and services produced within a country's borders in a year, regardless of ownership. It is used as one of the many indicators of the standards of living in a country, but there are limitations with this view.

Gilts: Government securities that are sold to the public are called Gilts.

Inflation: Inflation is a rise in the general level of prices of goods and services in an economy over a period of time.

KYC: Know your Customer or KYC details are a must to start any financial transaction with a mutual fund, bank account, broker account, etc.

Lock-in period: A period of time during which the investor is restricted from selling a particular investment.

Maturity date: Date on which the principal amount of a debt instrument or bond becomes due and payable in full.

Maturity Value: The amount the issuer agrees to pay out when the bond or term deposit certificate reaches its maturity date.

NAV: Net Asset Value (NAV) is the market value of mutual fund units. It is calculated each business day based on the value of the assets of the fund minus its liabilities, divided by the number of units outstanding.

Nomination: Facility of naming a person(s) to whom the assets may be distributed upon the death of the account holder.

Nationalisation: Transfer of ownership from private sector to public sector

Per capital income: Total national income of a country divided by its population in a specific period.

Redemption price: The price at which a mutual fund's units are redeemed or bought back by a fund.

SEBI: Securities and Exchange Board of India established under Securities and Exchange Board of India Act, 1992. It is the regulator for stock markets and market intermediaries like mutual funds, stock brokers, portfolio managers, etc.

Stock exchange: A market in which the securities of the government and public companies are traded. It provides the facilities for share brokers to trade company stocks and other securities.

Tariff barriers: All the restrictions on imports by a government in the form of taxes

Treasury bills (T-bills): A short-term debt instrument issued by the government with a maturity period of one year or less.

Zero Coupon Bond: A bond that is sold at a fraction of its face value. It does not, however, provide periodic interest payments but pays principal upon maturity

8.8 REFERENCES AND FURTHER READINGS

MacDougall, Curtis D. Interpretative Reporting (Sixth Edition). New York: Macmillan Publishing Company.

Debasis, Basu; Dalal, Sucheta (1993), The Scam: Who Won, who lost, who got away, Mumbai: South Asia Books

Norms of Journalistic Conduct (2005), New Delhi: Press Council of India.

8.9 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Economic and finance reporting are interrelated. However, economics is a broader subject and finance is a part of economics.

- 2) Economics is a very vast subject and deals with policy in regard to taxation, government spending, etc. Areas such as liberalization, privatization, globalization, poverty, human capital, rural development, employment, environment and sustainable development, etc. fall in the range of economics.
- 3) Financial news primarily deals with reporting share market, mutual funds, bullion prices, trading in consumer goods, company news, new products, etc.

Check Your Progress Exercise 2

- 1) The introduction of online trading in 1995 and the establishment of online depository of shares in 1996 were some important changes in stock markets. These led to buying and selling of shares through the Internet.
- 2) A financial reporter should have an excellent interpretative mind so that s/he can explain why stock market is falling, why people are taking more interests in fixed deposits than in stock market, and why mutual fund market is rising or falling, and so on.
- 3) State Bank of India, Indian Oil Corporation, ONGC, Coal India, etc.

Check Your Progress Exercise 3

- 1) Bombay Stock Exchange and National Stock Exchange of India.
- 2) Sensex is the market index consisting of 30 well-established and financially sound companies listed on BSE. Nifty is the market index consisting of 50 well-established and financially sound companies listed on National Stock Exchange of India (NSE).
- 3) Periodic reports of the Reserve Bank of India indicate the economic and financial health of the country.

Check Your Progress Exercise 4

- 1) Stocks of the leading and nationally known companies which offer a record of continuous dividend payments and other strong investment opportunities are known as Blue Chip Stocks.
- 2) Devaluation is a fall in the exchange rate that reduces the value of a currency in terms of other currencies.
- 3) Net Asset Value (NAV) is the market value of mutual fund units. It is calculated each business day, based on the value of the assets of the fund minus its liabilities, divided by the number of units outstanding.

UNIT 9 INDUSTRIALISATION AND URBANISATION

Structure

- 9.0 Introduction
- 9.1 Learning Outcomes
- 9.2 Concept of Industrialisation and Industrial Pollution
 - 9.2.1 Industrial Pollution
- 9.3 Industrialisation Special Economic Zone
- 9.4 Industrial Pollution – Environmental Impacts
 - 9.4.1 Industrial Air Pollution Episodes
 - 9.4.2 Industrialisation and Water Pollution
 - 9.4.3 Industrialisation and Soil pollution
 - 9.4.4 Industrialisation and Noise Pollution
- 9.5 United Nations Concept of Urbanisation
- 9.6 Problems of Urbanisation
- 9.7 Major Urban Development Projects in India
- 9.8 Role of Communication in Industrialisation and Urbanisation
- 9.9 Let us sum up
- 9.10 References and Further Readings
- 9.11 Answers to Check your Progress Exercises

9.0 INTRODUCTION

The process of industrialisation and urbanisation go hand in hand. First, let us see the growth process of both. When industrialisation started in the 19th Century, Cities developed and expanded around mining and other industrial areas. Along with industries, various other economic activities flourished, and the people migrated to the newly grown industrial areas in search of a better life. Thus industrialisation and urbanisation together expanded. Here we need to see whether industries a boon or a bane? Let us try to understand the various issues related to Industries from national and global perspectives and the implications of Industrialisation on the environment. Industrialisation began around the 17th Century in Britain and spread quickly to the other European countries. When we see from a historical perspective, we will discover that it is because of Industrialisation that nations have progressed from under-developed to developed of the current modern era. The process of industrialisation started in India immediately after we got independence. Indian Government started giving significance to industrialisation from the first five-year plan. It built heavy industries, started mining, developed infrastructure, built airports and ports, and developed industrial towns to proliferate. It was because of Industrialisation that our country got a stable economy. Since then, Industrial growth has helped us become self-reliant and has been a significant source of employment. However, in today's era, industries are looked upon not only as 'providers' but also as 'polluters'. They are being held responsible for the very survival of mother earth. How? and why? We will try to see this as we go through this Unit.

9.1 LEARNING OUTCOMES

After studying this Unit, you should be to:

- explain the concept of Industrialisation, Urbanisation and development;
- describe the environmental impacts of Industrialisation;
- define the concept of Special Economic Zones (SEZ);
- discuss the socio-economic impact of Industrialisation; and
- explain the concept of Urbanisation and Identify the problems associated with Urbanisation.

9.2 CONCEPT OF INDUSTRIALISATION AND INDUSTRIAL POLLUTION

There is no doubt that Industrialisation is the key to the progress of any country. Broadly, industries can be classified into different types depending upon their capital investment, i.e., Cottage Industries, Small Scale Industries and Large-Scale Industries. Irrespective of the type/size of the industry, one thing is fundamental to all industries: they are an essential source of employment and income for many rural and urban families. Indian Government gave importance to the industrialisation from the first five-year plan. It involved both public and private sectors in the development of industrialisation. Apart from that, the Government devised strategies to promote small industries at the village level through the Cottage industries commission and Khadi and Village industries commission. The industrialisation and economic contribution of manufacturing sectors didn't give expected results to the economy. Hence, Indian government opened up its economy to invite private players to enter specific sectors since the 1980s. It further privatized in 1991.

The Economic reform initiated in 1991 based on the principles of neoliberal economic policies also brought new reforms and showed the path to formulate new policies. The reforms also paved the way for forming the National Manufacturing Competitiveness Council in 2005, the National Strategy for Manufacturing in 2006, and the National Manufacturing Policy in 2011 (Soni and Bala Subrahmanya, 2020). The National Manufacturing and Investment Zones were created as part of the national manufacturing Policy (Subrahmanya 2014). The Government has removed controls and regulations over foreign trade and Foreign Direct Investment (FDI).

Reforms have also impacted differently in different sectors. The implemented reforms benefitted service sectors, and it became the largest sector in terms of Gross Value Added (GVA) at current prices. Approximately, GVA at current prices for the Service Sector is 92.26 lakh Crore INR in 2018-19. 54.40 per cent of GVA at current prices is the service sector from total India's GVA of 169.61 lakh crore INR. Concerning industry, it was 29.73 per cent which is 50.43 lakh crore. In comparison, the agriculture and allied sectors share only 15.87percent. I am further presenting the facts based on GDP composition. In the year 2017, Agriculture contributes 15.4 to GDP, 23 per cent from industry, and the most significant contribution comes from Services which was 61.5 per cent.

The contribution of agriculture and allied sector to the economy at current prices in 1950-51 was 51.81 per cent, the industry was 14.16 per cent, and the service was 33.25 per cent. There was a decline in the share of the agriculture and allied sector, which was 18.20 per cent in 2013-14. At the same time, the share of the service sector has improved to 57.03 per cent and the share of industry increase to 24.77 per cent.

(<http://statisticstimes.com/economy/sectorwise-gdp-contribution-of-india.php>).

The above facts showed the decline of the contribution of agriculture over the period. It was around 14 per cent of the country economy. But, still, it accounts for 42 per cent of the total employment (<https://tradingeconomics.com/india/gdp-from-agriculture>). The industrial sector did better than agriculture, but the service sector contribution to overall GDP is noteworthy. We learnt about industrialisation and economic development in this section. Now let us see the impact of industrialisation.

There are a few positive aspects of industrialisation which are given below:

A) Employment Opportunities

With heavy industrialisation come other developmental activities: power generation, mining, railways, road and air transport, post and telegraph services, etc. The industries and other projects employ millions of people who get an assured income. Thus, employment opportunities are provided to several people.

B) Changes in Economy

The developmental projects directly affect the economy as more income is generated through products, and many people get involved in the economic process. Also, the traditional kinds of activities are replaced by more advanced activities, and the resources are better utilized. Thus, providing better amenities and living conditions attracts more and more migratory populations, resulting in the development of cities and towns.

The Negative aspects of Industrialisation are given below:

1) Socio-Economic Impacts of Industrialisation

There is no doubt that the country has greatly benefited from industrial projects. However, it is unfortunate that while planning, the negative effects on the physical and social environment were not taken into account. We have seen in the previous sections how human beings has impacted the various components of our environment for the sake of industrialisation. But, it is also 'the man' who has been affected by the very same process of industrialisation in several different ways. The first and foremost issue related to any industrial project is that of land acquisition. Whenever a project is initiated, some agricultural/forest land is requisitioned for it. The population living there is displaced, and the affected persons have to leave their homes and profession and look for new jobs and a place to live. This brings profound changes in living conditions, employment patterns and social organisation amongst the local population. Some of the major socio-economic impacts of Industrialisation are:

2) Displacement of Population and Migration

The land is the first requirement for any developmental project. It isn't easy to find land that is not already inhabited or under cultivation in any overpopulated country. The first step in any project is clearing it of the human population. There are specific law to rehabilitate and resettled the displaced population with proper compensation (National Rehabilitation and Resettlement Policy,2007). Even though the displaced population properly rehabilitated and resettled, there are other consequences among displaced people. Most people migrate to already crowded cities and larger towns, where they can earn to sustain themselves. The main drawback of developmental activities is inadequate provision for rehabilitation of the affected population.

3) Change in Employment Pattern

While the developmental projects have created jobs but, on the other hand, have taken away the jobs of traditional workers and forced them to migrate to the cities. To take care of such anomalies, the Government and other organisations have to make efforts to provided employment to the jobless people who have been rendered unemployed in their areas. Dams, mining and irrigation projects displace large numbers of people from their land. Their rehabilitation thus becomes the prime concern for the Government.

4) Formation of Slums

Slums represent the worst type of environmental degradation that has become concomitant to urbanisation and industrialisation. Slums cause deterioration of the surrounding region and are a threat to human health. Some of the contributing factors causing the growth of slums are inadequacy or non-existence of civic amenities, failure to improve rules regarding house building and encroachment of public land, etc. Lack of basic facilities like water supply, electricity, sewers, stormwater drains, community baths and latrines and improved lanes in these areas pose health hazards and poor living conditions for the migrated population.

9.2.1 Industrial Pollution

As we mentioned in this Unit in the beginning, industrialisation does contribute to employment opportunities and improves the country's GDP. At the same time, most industries generate waste and contribute to environmental degradation due to the lack of proper disposal of industrial wastes. However, the quantities of wastes and the type of pollutants released vary significantly depends upon the industries.

Specific industries like mining and infrastructure development exploit natural resources. Industrialisation is responsible primarily for the exploitation of natural resources. It is the unthoughtful exploitation of natural resources which results in the depletion of biodiversity and resources and poses a danger to the very survival of humankind. The release of toxic/hazardous by-products during the manufacturing processes and the generation of large quantities of solid, liquid and gaseous wastes creates both treatment and disposal problems involving huge costs.

Industries are also classified depending upon the raw materials used, and the products made, for example:

- 1) Power plants: Coal-fired/Gas fired or Nuclear power plants. They are regarded as one of the largest contributors to environmental pollution.
- 2) Mining and Excavation Industries: These are also responsible for generating large quantities of hazardous waste and toxic pollutants.
- 3) Manufacturing Industries: Such as pulp and Papers mills, leather and textile industries, rubber and plastic industries.
- 4) Chemical and pharmaceutical Industries: Since these industries are involved in the manufacturing of acids, bases, pesticides, detergents and many other organic and inorganic compounds, they are responsible for releasing vast quantities of chemical wastes.
- 5) Agro-based Industries: These include sugar mills, brewing and distilling meat and dairy products, canning food processing etc. These industries are responsible for producing mainly organic wastes.

Besides, some other industries like health care, tourism, research laboratories, information technology, construction and other small commercial set-ups like photography, tailoring, dyeing, printing press, workshops etc., all contribute towards environmental pollution in either big or small way.

While on the one hand, industries are the critical sources for generating wealth and employment, on the other hand, they are responsible for any undesired consequences such as:

- Pollution of Air, Water and Soil
- Climate change
- Acid Rain
- Depletion of Natural resource
- Deforestation
- Degradation of Land
- Production of Hazardous wastes
- Adverse Health Effects
- Life-threatening Accidents

Thus, we can say that Industrialisation has led to environmental degradation of an undesirable degree. Developing countries like India are faced with challenges involving the protection of land, water, forest and air resources, provision of high-quality health standards and environmentally sound development initiatives. Thus, what we require today is ‘sustainable development’, which meets the needs of the present generation without compromising the needs of the future generations.

9.3 INDUSTRIALISATION – SPECIAL ECONOMIC ZONES (SEZ)

In the previous section, we introduced privatization and neoliberal economic policies. The concept of Special Economic Zones (SEZs) was an initiative of the Government of India, starting in 2000 as part of the Neoliberal economic policies,

to address the insufficient capital needs of the country by increasing the industrial capacity and putting India on an export-led growth path part of the neoliberal economic policies. The objective of SEZs was to address the infrastructural deficiencies, procedural and bureaucratic hassles, and have an inbuilt legal protection measure regarding monetary, trade, fiscal, taxation, tariff, and most importantly, labour policies. It started with some promising gains such as; employment opportunities for the local people and technology transfer from the more developed sectors to benefit the domestic industry. Thus, building SEZs was to have economic growth led by mainly private capital investment and the promotion of exports, widening the industrial base, research and infrastructure development.

Though the SEZs scheme promises an increase in employment and improvement in infrastructure, there is a contrasting view to this. Acquiring agriculturally productive land for setting up industries in SEZ is not an environmentally friendly decision. Besides, displacement of poor farm families and villages amounts to gross human rights violations. Thus, while on the one hand, SEZs promise a boost in the economy by promoting industrial growth, on the other hand, displacement of local farmers and acquisition of agriculturally productive land are some of the critical socio-economic and environmental concerns that need attention.

9.4 INDUSTRIAL POLLUTION- ENVIRONMENTAL IMPACTS

Industries have impacted our environment in a big way, be it the air we breathe, the water bodies, the soil or the habitat we live in. This section will see the impact of industrial pollution on air, soil, water bodies, and habitat.

9.4.1 Industrial Air Pollution episodes

Components of clean air that we breathe include Nitrogen (78.08%), Oxygen (20.95%), Argon (0.93%), Carbon Dioxide (0.033%) and other trace gases such as Neon, Helium, Methane, Krypton, Nitrogen Dioxide, Hydrogen. When the concentration of any trace gases increases beyond their acceptable level or when some harmful/ toxic gases or aerosols are introduced in the ambient air, they are termed pollutants.

Air pollution is caused due to both natural and anthropogenic activities. While on the one hand, volcanic eruptions and forest fires are some of the naturally contributing causes to air pollution. Industrial Revolution is one of the biggest manufactured causes of deteriorating air quality, particularly in areas inhabited by many people. Factories are dependent on fossil fuel burning extended release amounts of particulate matter and gases into the atmosphere. The release of these pollutants is responsible for an adverse health effect. Industries have been associated with many incidents of air pollution in the past that have been responsible for the morbidity and mortality of innocent people.

Environmental Impacts

Some severe environmental impacts produced by air pollutants of global and local magnitude are given below:

a) Greenhouse effect

A greenhouse refers to a glass chamber where plants are kept under controlled conditions. This greenhouse effect is seen in the environment also. Some gases present in the atmosphere perform a function similar to that of glass. The primary greenhouse gases are carbon dioxide, methane and nitrous oxide. These gases form a blanket over the earth's surface and allow the incoming short wave solar radiations to pass through and fall on the earth, thereby heating the earth's surface. However, these gases do not allow the long-wave radiations reflected from the earth's surface to escape into space. These reflected longwave radiations are re-emitted back to the earth's surface by the greenhouse gases, resulting in temperature build-up.

The most important greenhouse gases - carbon dioxide have increased by about 25% in the last 100 years. The concentration of CO_2 in pre-industrial times was about 280ppm and has increased to about 360ppm today. Fossil fuel combustion and deforestation are the main reasons for the increase in atmospheric CO_2 . The residence time of CO_2 is about 230 years. Based on the present-day modelling studies, it can be predicted that the eventual global warming of about 1.5 to 4.5°C will take place with the doubling of the CO_2 concentration. If this happens, it will have far-reaching consequences for life on earth.

Ozone Hole

Another essential global issue is stratospheric ozone depletion. The stratospheric ozone is commonly known as the 'Ozone Layer. It forms a protective covering over the earth's atmosphere and prevents harmful solar UV radiation. Worldwide monitoring of stratospheric ozone has shown that it has been decreasing for the past two decades or more. Globally averaged losses have been estimated to be about 5% since the middle of the 1960's with aggregate losses of 10% in spring and winter and 5% in summer and autumn over locations in Europe, North America and Australia. What is the cause of this ozone depletion? Significant increases in atmospheric concentrations of chlorine and bromine rich CFC's (chloro-fluoro-carbon compounds) released into the atmosphere are responsible for this depletion.

Acid Rain

Natural rainfall usually has a pH value between 5-6. Acid rain refers to rainfall with $\text{pH} < 5.0$. Oxides of sulfur are emitted into the atmosphere due to oxidation of fossil fuels, petroleum refining, copper smelting and cement manufacture. About 80% of oxides of sulfur coming from the burning of fossil fuels. Sulfur dioxide (SO_2), through series of chemical reactions, gets converted into SO_3 . This trioxide form of sulfur in reaction with water produces sulfuric acid (H_2SO_4), and this is the principal component (60-70%) of the acid rain. The conversion of sulfur dioxide into sulfuric acid is increased by metal ions like copper, lead and manganese and radicals like hydrocarbons and nitrogen oxides in the atmosphere and photochemical oxidation. Nitric acid (HNO_3) is the second most abundant component (30-40%), hydrochloric acid (HCl) the third and carbonic acid the fourth in acid rain. Acid rain can damage vegetation, corrode metals and building materials, weaken organic fibre and destroy aquatic organisms. Acid rain decreases the pH of water bodies. Certain toxic substances like aluminium, lead and mercury, which are insoluble otherwise, become soluble and can be life-threatening to

fish and other organisms. Prolonged exposure to acid rain can cause serious damage to building materials, including marble, limestone and mortar, as sulfates replace the carbonates in these materials.

9.4.2 Industrialisation and Water Pollution

The addition of any excess material (or heat) to water that renders it unfit /harmful for use by humans, animals, aquatic life or agriculture, or mere discharge into a larger water body is termed water pollution. Every use of water, be it, washing, irrigation, flushing of wastes, cooling, processing of some product, has been adding something or the other to water. Rivers and Oceans are often treated as the dumping grounds for human sewage and industrial wastes, many of which are highly toxic. Industries contribute to water pollution in the following manner:

- 1) **Thermal Pollution:** Water is often used as a cooling agent in many industrial processes, especially in power generating plants. This water carrying excess heat is generally released in nearby water bodies such as lakes, rivers etc., affecting aquatic life. The rise in temperature renders a water body unfit for the survival of fishes and other aquatic flora and fauna.
- 2) **Inorganic Chemical Pollution:** Many industries such as petrochemical, fertilizer factories, pulp, paper, textile, tanneries, distilleries, sugar and steel mills, drugs, fibres, rubber and plastic manufacturing Unit are responsible for discharging many kinds of organic and inorganic pollutants. The common industrial wastes discharged include metals (copper, lead, zinc etc), acids, alkalies, detergents, oil/grease, phenols cyanide, arsenic, chlorine and other toxic effluents that may bring about death or affect the reproductive, excretory, digestive or nervous systems. Some critical phenomena associated with water pollution are:
 - a) **Bio-accumulation**
Bioaccumulation occurs when a person is exposed to a substance over a while, accumulating higher levels of the substance in the body.
 - b) **Bio-magnification**
Bio-magnification occurs when organisms low on the food chain are consumed by organisms higher on the food chain, increasing the levels of the toxic agent in the predator as a result of concentration in the prey. For example, DDT, a pesticide that kills insects, is not easily degradable and remains in the environment for more than ten years once applied. This pesticide often biomagnifies as much as ten thousand times as it passes from one organism to another in the food chain.
- 3) **Organic Chemical Pollution:** Organic wastes released from mainly agro-based industries contribute heavily to biochemical oxygen demand (BOD) and chemical oxygen demand (COD) of water. These pollutants can render a water body stagnant. As the demand for oxygen increases, fishes and other aquatic flora and fauna are threatened. A critical phenomenon associated with water pollution is Eutrophication

Eutrophication refers to the enrichment of a water body by adding organic waste containing nutrients, mainly nitrates and phosphates. In the presence of oxygen, the aerobic bacteria in water bodies release nutrients

- 4) **Heavy Metals Pollution:** Serious health hazards have been reported due to the release of heavy metals like lead, cadmium and mercury in industrial wastes. Mercury is released as a by-product in the production of vinyl-chloride and from laboratories, hospitals, power plants and incinerators.
- 5) **Other Pollutants:** These include some chemical (i.e., disinfectants & their by-products) and biological (i.e. microorganisms) pollutants which lead to adverse health effects. Release of untreated municipal waste may build the concentration of E.coli and other pathogenic bacteria and viruses, capable of causing diseases like typhoid, cholera gastroenteritis etc. with symptoms of vomiting and diarrhoea. Thus, the Coliform index or Most Probable Number (MPN) is an essential indicator for measuring the E. coli count in water. The addition of disinfectants like chlorine in water, if added in large quantities, can also persist as 'Residual chlorine', which has an undesirable effect on fish & plankton mortality as well as in corrosion of pipes as it is a solid oxidizing agent.

9.4.3 Industrialisation and Soil Pollution

The main contributors to soil pollution are the vast quantities of solid wastes and chemicals. Industrial wastes like chemicals, fly ash and cinder (mainly from coal-based power plants and other industrial sectors), garbage, crop residues (from agro-processing units), paper, plastics and other packaging materials, discarded manufactured products or machines etc. all contribute towards soil pollution. Many fertilizers, pesticides / insecticides and pharmaceutical industries release chemicals directly into the soil or aerosols that ultimately settle down as soil pollutants. These chemicals may have severe implications for the species composition of plants and animals. These pollutants may inhibit the process of soil formation and may reduce the fertility of the soil.

Besides the soil pollution, industries have also been responsible for destroying the soil productivity by either encroaching an agriculturally productive land for industrial use, rendering it unfit for cultivation or by carrying out various industrial activities such as waste dumping, washing/cleaning, combustion and other such processes which ultimately lead to the release of undesirable pollutants into the otherwise productive and healthy soil.

9.4.4 Industrialisation and Noise Pollution

The presence of 'wrong sound' at the 'wrong time' in the 'wrong place' constitutes 'Noise'. Noise in an industry originates from processes causing impact, vibration/movements, friction and turbulence in air or gas streams. Several complex systems measure sound, but the best-known Unit of measurement is the decibel (dB). The decibel measures sound intensity or loudness. The human ear is known to be sensitive to vast range intensity from 0 to 180dB. While 0decible is the threshold of hearing, 140dB is the threshold of pain. By threshold, it implies the lowest intensity at which the stimulus gets perceptible. Some people feel discomfort even with a sound of 85 dB, whereas most do not feel discomfort with a sound of 115 dB. Pain is usually felt at 140 dB. Ordinary talk or discussion has the frequency range from 30-60 dB white noise produced by a jet plane at take-off may exceed 160 dB. The effect on man depends upon the frequency or pitch of the sound.

Noise is undesirable in any setup and has its ill effects. Some of the adverse effects of noise are:

Sleep interference, Effect on hearing, Effect on Communication, Other effects on health like affecting individuals' mentally and physically, **Effect on Work efficiency.** To prevent the population from adverse effects of noise, it is recommended that the noisiest parts of the factory should be kept as far as possible from the boundary of the site where workers/residents are present.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What are the socio-economic, environmental concerns of SEZs?

.....

.....

.....

.....

.....

2) List the various undesirable environmental consequences associated with the setting up of Industries.

.....

.....

.....

.....

.....

9.5 UNITED NATIONS CONCEPT OF URBANISATION

According to the United Nations Department of Economic and Social Affairs, Urbanisation is a complex socio-economic process that transforms the built environment, converting formerly rural into urban settlements, while also shifting the spatial distribution of a population from rural to urban areas. It includes changes in dominant occupations, lifestyle, culture and behaviour, and thus alters the demographic and social structure of both urban and rural areas. Today, the most urbanized regions include Northern America (with 82 per cent of its population living in urban areas in 2018), Latin America and the Caribbean (81 per cent), Europe (74 per cent) and Oceania (68 per cent). The level of urbanisation in Asia is now approximating 50 per cent. Meanwhile, Africa remains mostly rural, with 43 per cent of its population living in urban areas.

A major consequence of urbanisation is a rise in the number, land area and population size of urban settlements and in the number and share of urban residents compared to rural dwellers. Urbanisation is shaped by spatial and urban planning as well as by public and private investments in buildings and infrastructure. An

increasing share of economic activity and innovation becomes concentrated in cities, and cities develop as hubs for the flow of transport, trade and information. Cities also become places where public and private services of the highest quality are available and where essential services are often more accessible than in rural areas. In practice, urbanisation refers both to the increase in the population residing in urban areas and the associated growth in the number of urban dwellers, in the size of cities and in the total area occupied by urban settlements. Growth in the urban population is driven by an overall population increase and by the upward shift in the percentage living in urban areas. Together, these two factors are projected to add 2.5 billion to the world's urban population by 2050, with almost 90 per cent of this growth happening in Asia and Africa. Just three countries – India, China and Nigeria –are expected to account for 35 per cent of the growth in the world's urban population between 2018 and 2050. India is projected to add 416 million urban dwellers, China 255 million and Nigeria 189 million. Tokyo is the world's largest city with an agglomeration of 37 million inhabitants, followed by Delhi with 29 million, Shanghai with 26 million, and Mexico City and São Paulo, each with around 22 million inhabitants. Cairo, Mumbai, Beijing and Dhaka all have close to 20 million inhabitants. As per the UN report, the world's rural population has grown slowly since 1950 and is expected to reach its peak in a few years. The global rural population is now close to 3.4 billion and is expected to rise slightly and then decline to around 3.1 billion in 2050. Africa and Asia are home to nearly 90 per cent of the world's rural population. India has the largest rural population (893 million), followed by China (578 million).

9.6 PROBLEMS OF URBANISATION

The concept of Urbanisation started with the objective of providing facilities to meet the day to day needs in a better manner to improve the quality of living. Thus, better housing facilities with adequate water and electricity supplies and waste management facilities along with quality public services like schools, hospitals, transport etc. were developed in urban areas. However, due to better facilities in urban areas and a subsequent lack of basic infrastructure in rural areas, a large inflow of population started migrating to urban areas causing population explosion and overcrowding in urban areas. As a result, urban areas started feeling resource crunch as more and more were needed to meet these areas' ever-increasing demands.

An increase in the number of people due to either migration from nearby areas or rural areas in cities/towns/metros has led to a buildup of population or overcrowding in these areas. So much so that it has started building pressure on the available resources (i.e., land, water, transport, electricity, houses etc.), often leading to rampant construction activity and overexploitation of the present resources. Urbanisation, therefore, refers to mushrooming of urban dwellers who are constantly striving to attain more and more possessions, thereby leading to overexploitation of natural resources. With an ever-increasing population, urban areas are often faced with ever-increasing water, electricity, houses and other daily needs. Natural resources were overexploited, leading to the disappearance of the green cover. Mushrooming of high rise buildings and expanding network of roadways for rapidly multiplying vehicular traffic resulted in deterioration of air, water and soil quality. Thus, the concept of Urbanisation which started with the objective of improving the quality of life, ultimately led to the deterioration

of the living standard as the urban population was forced with more significant challenges of polluted air and water and newer threats of upcoming lifestyle diseases. The patterns of urbanisation in India has been marked by regional and interstate diversities, large scale rural to urban migration, insufficient infrastructural facilities, growth of slums and other allied problems. Some of the important problems of urbanisation faced in different parts of India are as follows:

Housing and Slums

There is an acute shortage of housing in urban areas, and much of the available accommodation is qualitative of sub-standard variety. This problem has worsened over the years due to the rapid increase in population, fast urbanisation, and inadequate addition to the housing stock. With large scale migration to urban areas, many find that the only option they have is to settle in slums. Slums are characterized by sub-standard housing, overcrowding, lack of electrification, ventilation, sanitation, roads and drinking water facilities. They have been the breeding ground of diseases, environmental pollution and many anti-social activities. India's slum population stands at approximately 40% slum dwellers and are reported to be maximum in Mumbai, Calcutta, Delhi, and Chennai.

Over Crowding

In major cities in India like Bombay, Calcutta, Pune and Kanpur, about 85% to 90% of the population lives in one or two rooms. In some homes, five to six persons live in one room. This over-crowding often results in rash behaviour, spreads diseases and creates conditions for mental illness, alcoholism and other social problems. One effect of dense urban living is people's apathy and indifference. Oscar Lewis' 'Culture of Poverty (1965) was an attempt to develop a model of the behaviour of the poor in a variety of cultural settings. It is a distinct way of life that develops among the lowest stratum in capitalistic societies in response to economic deprivation and inequality. Once people adapt to poverty, attitudes and behaviours initially developed in response to economic deprivation are passed on to subsequent generations through socialization.

Water supply, Drainage and Sanitation

Intermittent supply results in a vacuum created in empty water lines, which often suck in pollutants through leaking joints. Many small towns have no central water supply at all and are dependent on the wells. The drainage situation is equally bad. Because of the non-existence of a drainage system, large pools of stagnant water can be seen in the city even in the summer months. Removing garbage, cleaning drains and unclogging sewers are the main jobs of municipalities and municipal corporations in Indian cities.

Transportation and Traffic

The absence of planned and adequate arrangements for traffic and transport is another problem in urban centres in India. The majority of the people use buses and tempos, while a few use rail as a transit system. The increasing number of two-wheelers and cars make the traffic problem worse and add to air pollution. Moreover, the number of buses plying the metropolitan cities is not adequate, and commuters have to spend long hours travel.

Power Shortage

Power supply has remained insufficient in a majority of the urban centres in

India. The use of electrical gadgets has increased cities, and the establishment of new industries and the expansion of the old ones have also increased dependence on electricity. Conflict over power supply between two states often creates severe power crises for people in the city.

Pollution

Our towns and cities are major polluters of the environment. Several cities discharge 40 to 60 percent of their entire sewage and industrial effluents untreated into the nearby rivers. Urban industry pollutes the atmosphere with smoke and toxic gases from its chimneys. All these, increase the chance of diseases among the people living in the urban centres. According to UNICEF, lakhs of urban children die or suffer from diarrhoea and other water, air and vector-borne diseases because of poor sanitary conditions and water contamination. Most of the above-mentioned urban problems are migration and over urbanisation, industrial growth, apathy and inefficiency of the administration and defective town planning. Solutions to urban problems lie in the systematic development of urban centers and job opportunities, regional planning, and city planning, encouraging industries to move to backward areas and adopt a pragmatic Housing Policy. Towards the end of the Agenda for Sustainable Development in 2030, the share of the world's population living in urban areas is expected to reach 60 per cent. It is projected that by 2050, the world will be more than two thirds urban (68 per cent), roughly the reverse of the global rural-urban population distribution of the mid-twentieth century.

9.7 MAJOR URBAN DEVELOPMENT PROJECTS IN INDIA

To ensure that the benefits of urbanisation are shared and that no one is left behind, policies are drafted to manage urban growth needs and to ensure equal access to infrastructure and social services, focusing on the needs of the urban poor and other vulnerable groups for housing, education, health care, decent work and a safe environment.

It is predicted that by 2050 India will have the second-largest urban population in the world, and about 50% of India's population i.e., 814 million, is expected to live in urban areas. The present infrastructure and amenities in cities and towns are not adequate to address the expanding urbanisation process. Thus, several schemes have been launched by the Government to promote infrastructure development in the cities.

1) The Smart Cities Mission

The Smart Cities Mission is a major urban renewal program launched by the Government to develop and upgrade living conditions and infrastructure in selected 100 cities all over the country. The programme's objective is to modernize cities by providing core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment and application of 'Smart' Solutions. The programme was officially launched on the 25th of June 2016, and in the first phase, 20 cities will get funding for converting them into smart cities.

2) Atal Mission for Rejuvenation and Urban Transformation (AMRUT) Project

Atal Mission for Rejuvenation and Urban Transformation (AMRUT) was jointly planned and launched by the Governments to transform urban living conditions through infrastructure upgradation. AMRUT aims to transform 500 cities and towns into efficient urban living spaces over five years. Ministry of Urban Development has selected the five hundred cities with the help of state governments. The mission of AMRUT is to (i) ensure that every household has access to a tap with an assured supply of water and a sewerage connection; (ii) increase the amenity value of cities by developing greenery and well maintained open spaces (e.g. parks); and (iii) reduce pollution by switching to public transport or constructing facilities for non-motorized transport (e.g. walking and cycling).

3) Pradhan Mantri Awas Yojana (Urban) or Housing for All by 2022 Mission

The 'Housing for All by 2022' under the scheme of "Pradhan Mantri Awas Yojana – Housing for All (Urban)" launched by the Central Government aims to provide housing to all urban people by 2022. It provides central assistance to States and UTs for constructing houses to all eligible sections by concentrating on urban slums and economically weaker sections. Hence, slum rehabilitation and affordable housing to Economically Weaker Sections are the significant features of the project.

4) Heritage City Development and Augmentation Yojana (HRIDAY)

The HRIDAY scheme was launched for the holistic development of heritage cities. It aims to preserve and revitalize the unique character of heritage cities in India. Twelve cities—including Ajmer, Amaravati, Amritsar etc. are identified for the project.

5) Swachh Bharat Mission (SBM)

A major associated urban development programme is making India's urban centres clean. For this, the SBM targets to achieve 100 per cent scientific management of municipal solid waste in 4041 statutory towns/ cities in the country by 2019.

Besides the above initiatives, several associated development programmes like Urban transportation - to enhance public transport system including the Bus Rapid Transit Systems (BRTS), Industrial corridors, NHDP, Jawaharlal Nehru National Urban Renewal Mission, National Urban Livelihood Mission etc., aims to add infrastructure and other amenities to the urban areas.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Briefly describe any three major Urban Development Projects in India.

.....

.....

.....

2) What are the common problems of Urbanisation?

9.8 ROLE OF COMMUNICATION IN INDUSTRIALISATION AND URBANISATION

Communication can be defined as a process by which information is exchanged. One can express their thoughts/ideas/views and feelings with others through verbal/nonverbal expressions and writings. Communication is viewed as the means through which people of a community are linked together to achieve a common purpose of improving their living conditions. Communication is very critical to urban development as it is how the needs of the communities can be felt and catered to. Communication is therefore central to every human activity in the process of Urban development.

Communication should be relevant and practical to reach the various segments of society. The aims and objectives of the development projects must be properly communicated in the community. This should involve dialogue and discussions between the policymakers and the stakeholders. There should be regular communication regarding the needs and wants of the community members and ensure correct planning and implementation of the development policies. The participation of community members is central to promoting sustainable people-oriented urban development projects.

Communication would help the policy planners understand the problems confronting the communities and encourage them to find ways and means to address them through proper analysis and planning. A participatory collective approach involving community members would further strengthen the process of planning and implementation of urban development programs. Effective communication empowers the project participants in building confidence and trust amongst the members of the communities. The 74th Constitutional Amendment Act gave constitutional status to the urban local bodies and the act devolved various functions and functionaries to the municipalities and urban local bodies. It is the responsibility of the media to strengthen urban local bodies to improve governance in urban areas.

9.9 LET US SUM UP

After going through this Unit, you would have got an idea about the concept of industrialisation and also how development and environmental degradation go hand in hand. You have studied how each of the industrial set up (i.e. power plants, mining and excavation industry, manufacturing industry, chemical and

pharmaceutical industry, agro-based industry etc.) are responsible for the various undesirable consequences such as: air, water and soil pollution, climate change, acid rain, natural resource depletion, deforestation, land degradation and adverse health effects, lifestyle diseases and life-threatening accidents. You have studied some cases of pollution episodes and life-threatening industrial accidents in India and abroad in the past. These set an example for all of us to be careful while planning any industrial/development project in the future and considering all possible environmental consequences that a project can have. This would ensure sustainable development for our present and future generations. You have also studied some of the global environmental impacts and the adverse effects of water, air, soil and noise pollution on human health and other living organisms. You have also studied the various socio-economic impacts that Industrialisation has brought with it, be it positive and negative ones.

As you have seen, urbanisation is an ongoing phenomenon that is very difficult to capture through any single approach or analysis, especially in India. In this Unit, we have studied different aspects of urbanisation, approaches to study urbanisation and the problems and consequences of urbanisation. Some major Urban Development Projects in India were also discussed, which aim at providing housing and basic infrastructure facilities to the ever-increasing population of cities and towns.

9.10 REFERENCES AND FURTHER READINGS

Myers, N. 2001. Environmental Refugees: Our Latest Understanding. *Philosophical Transactions of the Royal Society B*: 356: 16.1-16.5.

Davis D.L, Bell M.L, Fletcher T. 2002. A look back at the London smog of 1952 and the half-century since. *Environment Health Perspective*. 110 (12), December.

Helfand W.H, Lazarus J, Theerman P. 2001. Donora Pennsylvania: environmental disaster of the 20th century. *American Journal of Public Health*. 91: 553-554.

Glasser M., Greenburg L. and Field F., 1967: Mortality and Morbidity during a Period of High Levels of Air Pollution. *Arch. Environmental Health*. 15, 684-694.

ICMR Bulletin. 1997. Air Pollution: Problems and Perspectives. Vol. 27 (6).

Gupta K.M. 2009. Environment and Ecology. 4th Edition.

Hilgenkamp Kathryn. 2006. Environmental Health-Ecological Perspectives.

World Urbanisation Prospects 2018. 2019. Highlight. United Nations, Department of Economic and Social Affairs, Population Division.

9.11 ANSWERS TO CHECK YOUR PROGRESS EXERCISE

Answers to Check Your Progress Exercise 1

- 1) Displacement of local farmers and acquisition of agriculturally productive land, loss of income generation mainly of farmers.

- 2) Pollution of Air, Water and Soil; Climate change; Acid Rain; Natural resource depletion; Deforestation; Degradation of Land; Production of Hazardous wastes; Adverse Health Effects; Life-threatening Accidents

Answers to Check Your Progress Exercise 2

- 1) Positive: Changes in Economy and employment opportunities. Negative: Displacement of local Population, Migration, formation of slums.
- 2) Environmental Refugees is a new class of people who are displaced from their native areas due to certain environmental catastrophes such as rising sea levels, droughts, desertification, deforestation, weather-induced flooding and other serious natural disasters due to climate change. These are people who can no longer gain a secure livelihood in their homelands because of environmental problems, together with associated problems of population pressures and profound poverty.



